EEVblog Electronics Community Forum

Electronics => Projects, Designs, and Technical Stuff => Topic started by: Jay_Diddy_B on September 08, 2013, 01:51:52 am

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Title: **Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **September 08, 2013, 01:51:52 am**

Hi group,

A while a go I shared my design for a small electronic load that was limited to dc operation only. I am in the process of design and construction of a dynamic load. The dynamic load steps the load current so that the transient response of the power supply being tested can be observed.

I have decided to keep the design simple using readily available components. I have also focused on performance of the analog circuitry.

Features:

0-5A maximum continuous current
0-5A pulsed current at 330Hz

Maximum dissipation around 40W (There is no protection circuits in the design)

LTspice was used for circuit modelling. I have attached the LTspice model in a zip file attached to this post.

The prototype was constructed on a double-sided board made on an LPKF milling machine.

LTspice schematic:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59887;image)

LTspice results

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59889;image)

Risetime:
(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59891;image)

Falltime:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59893;image)

Construction:

Real Schematic

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59885;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59895;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59897;image)

Note: The MOSFETs will be bolted to a heatsink

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59899;image)

Hardware Tests:

The current was measured with a Tektronix TCP202 current probe

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59901;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59903;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59905;image)

I will post some additional pictures when I complete the mechanical part of this project.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **September 08, 2013, 02:26:11 am**

Thanks a lot JDB, I've been waiting for this kind of circuit for long time. :-+

YES !!! Finally, an affordable dynamic load circuit using just common jelly bean components ! :clap: :clap: :clap:

Btw, is it true at the statement that any serious bench/lab power supply should be tested with "dynamic" load instead of just "static" load ?

Ok, now I'm going to observe & learn this circuit, and for sure I will come up with tons of noob questions to bug you :-[, hope you don't mind.

;D

Again, thank you. :-+

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **September 08, 2013, 02:48:34 am**

BravoV,

A dynamic load can be used to generate a disturbance in a power supply's control loop.

The response to a step load change give an indication of a power supplies stability margins.

The Dynamic load is very easy and fast to use.

Serious power supply work in done with a Frequency Response Analyzer. This is a complicated and expensive piece of equipment that will display a Bode plot of the power supply's control loop. Hint: Google "Ridley Engineering", "Venable Instruments" or "Omicron FRA"

Playing with the LTspice model is the best way to figure out how this circuit works.

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **dannyf** on **September 08, 2013, 03:18:27 am**

"show some pictures for a user interface. This device needs to function for more than 20 years"

paintings or carvings, like those by ancient cavemen - they have survived millions of years.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **September 08, 2013, 02:01:34 pm**

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dannyf,

The user interface is two knobs. Knobs have been around for a hundred years and will still be current technology in the next 20 years. ;D

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **September 08, 2013, 02:05:33 pm**

Hi,

I am going to share the ac analysis that was used to ensure that the electronic was stable.

LTspice model

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59968;image)

The LTspice model includes a source inside the control loop. The source injects a disturbance into the loop. The signal is measured on either side of the source V(A) and V(B).

The .meas directives in the LTspice model is a network analyzer :D

To see the Bode plot run the model in the normal way. The model will run 21 times, with different frequencies being injected into the loop.

After the last simulation is complete, click on the schematic, then click on View menu. Select SPICE error log. Right click in this window and select plot step'ed meas data. When asked 'Shall I write these as complex data?' click yes.
Right click in the new window, select add trace. Select gain from the list. This will display the Bode plot.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59959;image)

Damping Network R7 C2

The damping network R7 and C2 to compensate for the lead inductance to the power supply under test. If I remove the circuit by changing R7 from 2.2 Ohms to 2.2 MOhms:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59961;image)

Then I get the following result:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59963;image)

If I increase the Lead Inductance to a value greater than 1.8uH the circuit will oscillate.

If I add the damping circuit the circuit is stable.

I have attached a zip file with the LTspice model.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **September 09, 2013, 06:47:39 am**

JDB, few early questions and for sure more to come while I'm digesting this circuit. >:D

1. Why 330 Hz ? Is it because of your test at specific condition that you currently need ? What if that oscillator section is modded into adjustable frequency ? What range of the frequency sweep that is allowed without affecting the loop stability ?

2. If I replace the square wave oscillator say with wave gen that capable of producing various pulse shapes like sawtooth, triangle, shark fin, sine and etc, again, as above, will it affect the loop stability ?

3. I see you use two mosfets, what are needed to change/alter on the circuit if I use 4 mosfets (of course with it's own respective driver) ?

4. I have tons (bought from industrial surplus) National/TI LF411 jfet opamp, is it ok to use it instead of TL074 without the need to alter other components value, again, just worry about its stability, cause honestly I'm not very firm at gripping at how exactly to stabilize it yet, pardon, I'm still learning. :-[

5. Tried the simulation at your 1st circuit on ltspice, adjusted with lower offset and lower vref and I got this overshoot at the simulation, as you can see its almost at 100% more than the programmed current. Is this just at simulator ? or something else ? or just ignore it ? Check below pic.
(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59988;image)

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **September 09, 2013, 11:51:06 am**

BravoV,

Quote from: BravoV on September 09, 2013, 06:47:39 am

> JDB, few early questions and for sure more to come while I'm digesting this circuit. >:D
>
> 1. Why 330 Hz ? Is it because of your test at specific condition that you currently need ? What if that oscillator section is modded into adjustable frequency ? What range of the frequency sweep that is allowed without affecting the loop stability ?

The reason 330Hz was chosen is that most power supplies have more than 1kHz of bandwidth. An optimized switching supply can have a bandwidth as high 125kHz. At 1kHz I wanted to be able to a couple of cycles. I then used resistors values and capacitors that I had.
When I am using my HP6060A load I normally select 100Hz or 1KHz.

Quote from: BravoV on September 09, 2013, 06:47:39 am

> 2. If I replace the square wave oscillator say with wave gen that capable of producing various pulse shapes like sawtooth, triangle, shark fin, sine and etc, again, as above, will it affect the loop stability ?

The U3 -input (on the LTspice model) is a summing node. You can add the output from your signal generator, through a 100K resistor, to this input. You can then use all the waveforms in signal generator.

Quote from: BravoV on September 09, 2013, 06:47:39 am

> 3. I see you use two mosfets, what are needed to change/alter on the circuit if I use 4 mosfets (of course with it's own respective driver) ?

You can use 4 MOSFETs, I just wanted to show more than one. I also wanted to limit my design to using a single quad op-amp. You just replicate the MOSFET, sense resistor, op-amp circuit. I would recommend 1 MOSFET for each 25W of dissipation. This arrangement forces the MOSFETs to share the current.

Quote from: BravoV on September 09, 2013, 06:47:39 am

> 4. I have tons (bought from industrial surplus) National/TI LF411 jfet opamp, is it ok to use it instead of TL074 without the need to alter other components value, again, just worry about its stability, cause honestly I'm not very firm at gripping at how exactly to stabilize it yet, pardon, I'm still learning. :-[

The LF411 is fine. It has similar GBW and slew rate.

Quote from: BravoV on September 09, 2013, 06:47:39 am

> 5. Tried the simulation at your 1st circuit on ltspice, adjusted with lower offset and lower vref and I got this overshoot at the simulation, as you can see its almost at 100% more than the programmed current. Is this just at simulator ? or something else ? or just ignore it ? Check below pic.
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59988;image)

I haven't evaluated this fully, but I suspect it due the power supply sequencing in the model.

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **sorin** on **September 09, 2013, 02:20:37 pm**

I think that is something wrong with "dynamic load AC ANALYSIS.zip " please check it
when I run the stimulation the results are much different from yours
and don't have sense

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **September 09, 2013, 03:46:44 pm**

Sorin,

I have corrected the attachment in the original post.

There were two issues:

1) V2 should have been -0.1V instead of +0.1V

2) The 2.2Ohm 2.2uF damping network was missing.

The schematic shown are correct.

Thank you for pointing this out :-+

Jay Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **September 10, 2013, 08:14:13 am**

JDB, thanks for the concise replies, really help me a lot. :-+

> Quote from: Jay_Diddy_B on September 09, 2013, 11:51:06 am
> > Quote from: BravoV on September 09, 2013, 06:47:39 am
> > 5. Tried the simulation at your 1st circuit on ltspice, adjusted with lower offset and lower vref and I got this overshoot at the simulation, as you can see its almost at 100% more than the programmed current. Is this just at simulator ? or something else ? or just ignore it ? Check below pic.
> >
> > I haven't evaluated this fully, but I suspect it due the power supply sequencing in the model.

Looking forward to see the result after your evaluation. :-+

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **September 10, 2013, 08:58:14 am**

BravoV,

The current overshoot that you observed is the damping network Capacitor being charged when the V3 is ramping up at the start of the simulation. This should not be a problem because the damping network is in parallel with the output capacitor of the power supply being tested.

The damping network is key to getting a large bandwidth.

It is damping a resonance that occurs between the output capacitance of the MOSFET and the lead inductance used to connect the load to the power supply under test.

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **September 11, 2013, 05:05:37 am**

> Quote from: Jay_Diddy_B on September 09, 2013, 11:51:06 am
> The reason 330Hz was chosen is that most power supplies have more than 1kHz of bandwidth. An optimized switching supply can have a bandwidth as high 125kHz. At 1kHz I wanted to be able to a couple of cycles. I then used resistors values and capacitors that I had.
> When I am using my HP6060A load I normally select 100Hz or 1KHz.

I'm not very sure about this, the switching supply bandwidth is not the same as switching frequency, right ?

> Quote from: Jay_Diddy_B on September 10, 2013, 08:58:14 am
> The current overshoot that you observed is the damping network Capacitor being charged when the V3 is ramping up at the start of the simulation. This should not be a problem because the damping network is in parallel with the output capacitor of the power supply being tested.

Noted, and yes, I forgot the output cap of the V3 which is always there at every PSU, thanks.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **October 12, 2013, 11:49:38 pm**

Hi,
I got some time this weekend to work on my Dynamic Load project. So here is an update.

The case that I am using is a aluminium extrusion. The case slots for mounting the board. I don't know who made the case because I reused one that I had found at a junk shop.

I made a new board for the project. The changes were mainly mechanical. I added some Keystone 594K and 593K 9V battery holders. I also added 3mm to the length of the board, because I had forgot to include the dimension of the plastic molding around the front panel. I also adjusted the spacing of the BNC, pots and binding posts to fit the case. There is only just enough space on the front panel.

SMD side of the board:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=63295;image)

Thru-hole side of the board:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=63297;image)

Front Panel:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=63299;image)

Here are some test results:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=63301;image)

This image shows the current rise time. The upper trace is from the monitor port on the dynamic load. The lower trace is from a Tektronix TCP202 current probe.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=63303;image)

This image shows the current fall time. The upper trace is from the monitor port on the dynamic load. The lower trace is from a Tektronix TCP202 current probe.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=63305;image)

This image shows a few cycles of the current waveform. The upper trace is from the monitor port on the dynamic load. The lower trace is from a Tektronix TCP202 current probe.

For comparison here is the Tektronix TCP202 DC current probe monitoring a HP50501B in a 6050A mainframe, under the same conditions:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=63307;image)

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **October 13, 2013, 07:09:05 am**

Thanks for the update, the case looks beautiful and the results are identical with that current probe result. :clap:

Btw, abit nitpicking ^-^ on the two 9 volt batteries, are they secured only at it's terminal ?

Worry they may detached them self from the terminals if there is a hard bump on the case at the front, and somehow at worst case scenario the metal body of the battery shorted out the (+) terminal that is near behind it vs the screw head. A padding at the battery's bottom to push them firmly against those terminals maybe ? Just a thought.

Title: **Re: Dynamic Electronic Load Project**
Post by: **grenert** on **October 13, 2013, 03:36:39 pm**

Quote from: Jay_Diddy_B on October 12, 2013, 11:49:38 pm

> The case that I am using is a aluminium extrusion. The case slots for mounting the board. I don't know who made the case because I reused one that I had found at a junk shop.

Very cool project, thanks for sharing it with us!
I think your case looks like a Hammond:
http://www.hammondmfg.com/1455V2.htm (http://www.hammondmfg.com/1455V2.htm)

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **October 13, 2013, 06:00:46 pm**

Hi,

I thought it might have been one of those Hammond cases, and I would have probably selected one, if I didn't find the surplus ones.

Here is a picture of the extrusion, the outside dimensions are 5.26 x 1.46 inches.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=63427;image)

Here is a picture of the completed project.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=63429;image)

Quote from: BravoV on October 13, 2013, 07:09:05 am

> Btw, abit nitpicking ^-^ on the two 9 volt batteries, are they secured only at it's terminal ?
>
> Worry they may detached them self from the terminals if there is a hard bump on the case at the front, and somehow at worst case scenario the metal body of the battery shorted out the (+) terminal that is near behind it vs the screw head. A padding at the battery's bottom to push them firmly against those terminals maybe ? Just a thought.

I will probably put a little double-sided foam pad to secure the batteries. Good Idea !!

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Dave Turner** on **February 20, 2014, 11:00:50 pm**

Nice and elegant. How do you determine the pot settings when using the unit?

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **February 21, 2014, 12:23:07 am**

Start at lowest position while watching the scope, and slowly crank the pot up ?

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **June 16, 2014, 02:00:44 am**

Hi group,

There has been a lot of interest in the forum on building constant current loads. One of the frequent questions that is asked is how do I stop my load from oscillating? or how do I choose the components around the op-amp to obtain stability. Other popular questions are how does the MOSFET gate capacitance impact my design.

I am going to share some analysis on a constant current load using LTspice. The analysis is done in the frequency domain. The goal is get reasonable bandwidth while being stable. For stability the phase shift has to be less than 180 degrees when the gain is greater than 1 (0dB). The loop bandwidth and phase margin are figures of merit for a control system.

Since the gain of a control loop is the product of the gains of the individual stages, we can build the control loop from the building blocks used in the circuit.

Power Stage

I could start by trying to model the power stage like this:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97815;image)

And I get the following result, which is useless:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97817;image)

The reason that the result is useless, is that the MOSFET has not been biased to the correct operating point.

The easy way to correct this is add an automatic Bias generator. This will servo the gate voltage to the correct operating point:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97819;image)

The result from this circuit are:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97821;image)

I can also reduce the MOSFET to a voltage controlled current source and its Gate-Drain and Gate-Source Capacitors:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97823;image)

If I model the correct values of transconductance, Crss and Ciss I get the same results. I do not need a bias circuit, because this model has no gate threshold voltage:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97825;image)

I now have a model that I can explore the effects of the gate capacitors directly. The DC gain is determined by the value of the source and the transconductance of the MOSFET. There is a pole formed by the gate resistor and input capacitance of the MOSFET.

Op-Amp Circuit

An op-amp is used to stabilize the load current. Typical this circuit it used:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97827;image)

The results show a single pole. It important that the op-amp has sufficient gain bandwidth product (GBW) so that the op-amp does not limit the bandwidth.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97829;image)

The circuit can also be reduced to this:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97831;image)

The results for this circuit are:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97833;image)

You can see that the real circuit, with the op-amp, has the same characteristics as the simplified model. The characteristics are determined by the resistor and the capacitor.

Open Loop Model

The two previous models can be placed in series to obtain the control loop response (oops!! replace R5 and L1 with a short :palm:):

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97835;image)

The loop model response is:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97837;image)

This is great and the control loop is stable. What would happen if there is some inductance in the leads between the load and the power supply being tested?

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97839;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97841;image)

The results indicates that the circuit is very close to oscillating. There is very little phase margin when the gain is 0dB. We could reduce the gain, but this would also reduce the bandwidth of the control loop.

We could add a damping circuit as shown in this model:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97849;image)

The result is:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97851;image)

The potential for oscillations resulting from inductance on the leads to the load has been significantly reduced.

Closed loop Modelling

The loop gain can be measured with the control loop closed, by placing the disturbance source in the feedback path. The signal can be measured on either side of the disturbance source to determine the loop gain.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97843;image)

The results of the closed loop model are:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97845;image)

To be continued....

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **June 16, 2014, 02:01:18 am**

Continuing...

Lead Inductance can be added to the closed loop model:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97853;image)

An the results are similar to the open loop model, the circuit is on the edge of oscillation:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97855;image)

An RC damping Circuit can be added to the model:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97857;image)

And the stability margins are restored:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97859;image)

I hope this analysis give some insights in to how to design a stable load.

May all your loads be stable !!

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **June 16, 2014, 03:40:31 am**

Quote from: Jay_Diddy_B on June 16, 2014, 02:01:18 am

> May all your loads be stable !!

LOL, thanks for the updates.

As a noob, after reading these excellent additions, suddenly I feel like an expert now. >:D :palm:

Title: **Re: Dynamic Electronic Load Project**
Post by: **kt315** on **June 17, 2014, 03:27:21 am**

Jay_Diddy_B,

Thanks a lot for putting in the effort to explain your analysis. It is by far the most clear explanation I've seen so far. It is the most useful explanation I have seen so far ... :-+

Could you explain in a bit more details please how you read the last two charts of your post. Specifically, how do you determine if the circuit is stable.

If I read it right, the unity gain of the loop is at 30 kHz. But in both cases (before and after dampening is added) the phase shift seems to be pretty far away from 180.
Or do I read it wrong?

Thanks.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **June 17, 2014, 09:00:16 am**

Quote from: kt315 on June 17, 2014, 03:27:21 am

> Jay_Diddy_B,
>
> Thanks a lot for putting in the effort to explain your analysis. It is by far the most clear explanation I've seen so far. It is the most useful explanation I have seen so far ... :-+
>
> Could you explain in a bit more details please how you read the last two charts of your post. Specifically, how do you determine if the circuit is stable.
>
> If I read it right, the unity gain of the loop is at 30 kHz. But in both cases (before and after dampening is added) the phase shift seems to be pretty far away from 180.
> Or do I read it wrong?
>
> Thanks.

Hi,
I have zoomed in on the area that we are interested in. Here is the results without the damping network:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97991;image)

I have used arrows to indicate the phase margin and the gain margin. The phase margin is a very good 84 degrees. The gain margin is low, only 3dB.

If I add the damping network. I get the following result:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97993;image)

I have the same phase margin 84 degrees, but the gain margin has increased to a massive 52dB.

So why is this problem? :-//

Both of these loops are stable. The issue is that part of the control loop, the transconductance of the MOSFET varies with the operating point. The gain increases as the current increases. If I modify my model to this:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97995;image)

Here I am stepping the load current from 0.5A to 5A. I get this result:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97997;image)

The high current result is unstable, the gain is 3dB when the phase angle is 0 degrees.

If I switch to the time domain. In this model I am ramping the load current from 0 - 5A:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97999;image)

I get this result:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=98001;image)

You can see the control loop breaks into oscillation when the current is greater than 1.9A.
Note, that I had to add a little ripple, just 10mV, to disturb the control loop to get the oscillation to show up.

If I add the damping network I can sweep the current from 0-5A and there are no oscillation as predicted in the frequency domain. (There is nothing to see, so I have not included the results).

An alternative solution to the damping network, is to reduce the loop gain. If I increase the capacitor by factor 10, to 2200pF, I get this result:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=98003;image)

This is a very stable system, but the limited bandwidth, would prevent the circuit being useful for transient load tests.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **kt315** on **June 17, 2014, 11:23:38 pm**

You are my hero. Thanks a lot.

Title: **Re: Dynamic Electronic Load Project**
Post by: **RichardD** on **January 15, 2015, 11:28:09 pm**

Hi - can you comment on why you used an inverting op amp configuration? Typically you see electronic load circuits that use a non-inverting op amp configuration, so just wondering if there were any particular advantages to the non-inverting configuration (e.g., stability, easier to configure the summing node, etc.)?

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **January 16, 2015, 01:43:44 am**

Quote from: RichardD on January 15, 2015, 11:28:09 pm

> Hi - can you comment on why you used an inverting op amp configuration? Typically you see electronic load circuits that use a non-inverting op amp configuration, so just wondering if there were any particular advantages to the non-inverting configuration (e.g., stability, easier to configure the summing node, etc.)?

The main reason for using the inverting configuration was it allows the dc reference and the pulsed reference to be easily summed together.

The main disadvantage of the circuit is that two power supplies are required. The op-amps are operating with their inputs and outputs between the rails. The allows a very large selection of op-amps to be used in the circuit.

The non-inverting configuration, when operated with a single supply, does not require rail-to-rail inputs, but it does require op-amps that work with their inputs at the negative supply voltage. These include LM324, LM358 etc.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **January 21, 2015, 03:20:30 pm**

Quote from: AcHmed99 on January 21, 2015, 01:44:07 pm

> Hey Jay your project is listed over on DP. ;)
>
> http://dangerousprototypes.com/2015/01/20/diy-dynamic-electronic-load/ (http://dangerousprototypes.com/2015/01/20/diy-dynamic-electronic-load/)

Well deserved ! :clap: :-+

Title: **Re: Dynamic Electronic Load Project**
Post by: **diyaudio** on **January 21, 2015, 03:31:27 pm**

Quote from: Jay_Diddy_B on June 17, 2014, 09:00:16 am

> Quote from: kt315 on June 17, 2014, 03:27:21 am
>
> > Jay_Diddy_B,
> >
> > Thanks a lot for putting in the effort to explain your analysis. It is by far the most clear explanation I've seen so far. It is the most useful explanation I have seen so far ... :-+
> >
> > Could you explain in a bit more details please how you read the last two charts of your post. Specifically, how do you determine if the circuit is stable.
> >
> > If I read it right, the unity gain of the loop is at 30 kHz. But in both cases (before and after dampening is added) the phase shift seems to be pretty far away from 180.
> > Or do I read it wrong?
> >
> > Thanks.
>
> Hi,
> I have zoomed in on the area that we are interested in. Here is the results without the damping network:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97991;image)
>
> I have used arrows to indicate the phase margin and the gain margin. The phase margin is a very good 84 degrees. The gain margin is low, only 3dB.
>
> If I add the damping network. I get the following result:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97993;image)
>
> I have the same phase margin 84 degrees, but the gain margin has increased to a massive 52dB.
>
> **So why is this problem? :-//**
>
> Both of these loops are stable. The issue is that part of the control loop, the transconductance of the MOSFET varies with the operating point. The gain increases as the current increases.
> If I modify my model to this:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97995;image)
>
> Here I am stepping the load current from 0.5A to 5A. I get this result:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97997;image)
>
> The high current result is unstable, the gain is 3dB when the phase angle is 0 degrees.
>
> If I switch to the time domain. In this model I am ramping the load current from 0 - 5A:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97999;image)
>
> I get this result:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=98001;image)
>
> You can see the control loop breaks into oscillation when the current is greater than 1.9A.
> Note, that I had to add a little ripple, just 10mV, to disturb the control loop to get the oscillation to show up.
>
> If I add the damping network I can sweep the current from 0-5A and there are no oscillation as predicted in the frequency domain. (There is nothing to see, so I have not included the results).
>
> An alternative solution to the damping network, is to reduce the loop gain. If I increase the capacitor by factor 10, to 2200pF, I get this result:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=98003;image)
>
> This is a very stable system, but the limited bandwidth, would prevent the circuit being useful for transient load tests.
>
> Regards,

> Jay_Diddy_B

@Jay_Diddy_B

Thanks for this,studying compensation networks myself and these these simulations looks pretty good.

Title: **Re: Dynamic Electronic Load Project**
Post by: **scopeman** on **March 16, 2015, 02:26:07 am**

Hi Jay,

Do you have a PCB layout file or Gerbers you can post for this circuit?

Thanks,

Sam
W3OHM

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **March 16, 2015, 02:53:55 am**

Quote from: scopeman on March 16, 2015, 02:26:07 am

> Hi Jay,
>
> Do you have a PCB layout file or Gerbers you can post for this circuit?
>
> Thanks,
>
> Sam
> W3OHM

Sam,

Here are the Gerbers and the latest version of the schematic.

This is a two sided layout. There are a few vias that connect the two sides.

The board was designed to fit the extruded case that I found at my local junk shop.

73,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **March 16, 2015, 03:16:33 am**

Hi,

The Gerbers should look like this:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=141992;image)

And

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=141994;image)

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **November 01, 2016, 03:03:12 pm**

Quote from: Jay_Diddy_B on September 08, 2013, 01:51:52 am

> ... <snip>....
>
> Features:
>
> 0-5A maximum continuous current
> 0-5A pulsed current at 330Hz

Jay, I have this 500V 46A "linear" mosfet IXTN46N50L (photo (https://www.eevblog.com/forum/chat/what-did-you-buy-today-post-your-latest-purchase!/?action=dlattach;attach=221228;image)) , datasheet (HERE (http://ixdev.ixys.com/DataSheet/89945a33-9e28-4242-b812-008abd738156.pdf)).

It has a whopping gate capacitance (Ciss) at 7000 pF, just wonder if the circuit needs a major revision just to make it work as pulsing dummy load ? Say at "reasonable" rise & fall time for working at "common" power supply types. Also when running at the static continuous current, does need modification too ?

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **November 02, 2016, 02:04:47 am**

BravoV,

Let start by coming up with a specification. That MOSFET on a CPU cooler should handle around 200W. So let us set the maximum current at 20A.

If we choose a voltage drop on the shunt of 200mV, we have 4W of dissipation in the shunt.
20A/0.2V = 10m?

We can make the sense resistor using ten 100m? resistors in parallel.

AC Analysis

We repeat some of the AC analysis that was performed in this message.

https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/msg462562/#msg462562
(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/msg462562/#msg462562)

We can the value of transconductance from this graph on the datasheet. Using a value of 12 will give us a little margin.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266914;image)

The capacitance varies as a function of the drain source voltage. This is shown on the datasheet like this:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266916;image)

The values at Vds =3V and Vds=15V will be used in the analysis.

Small signal SPICE model

A small signal SPICE model can be built using the information derived from the datasheet:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266918;image)

The results show that the bandwidth of the output stage is a respectable 190 kHz.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266920;image)

Repeat the analysis at Vds =15V

Vds=15V Model

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266922;image)

Vds=15V Results

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266924;image)

The 3dB point is slightly higher in frequency at 240 kHz.

Gate Drive Requirements

The transfer characteristic of the MOSFET are shown in this graph from the datasheet:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266926;image)

The graph reveals that we will need to able to provide at least 12V of gate drive. So the op-amps we use will be powered from +15V and -5V rails.

We can add the transfer characteristics to the small signal model by putting a dc voltage in series with the gate:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266928;image)

To be continued ....

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **November 02, 2016, 02:16:50 am**

Complete Load
The error amplifier can be added to the model:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266930;image)

Note: the model includes an inductor to representing the wiring to the power supply. An RC damping network has been added, this is explained earlier in this thread.

The model can be tested at the 3V operating point:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266932;image)

The model can be tested at the 15V operating point:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266934;image)

Bode plots

The model can be modified to look at the small signal behaviour, by inserting a disturbance in the feedback path. Plot V(a)/V(b) to display the control loop gain.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266936;image)

Bode plot for 3V

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266938;image)

Bode plot for 15V

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266940;image)

The control loop has about 25 kHz of bandwidth. This is fairly conservative. I have not built this circuit, so I am giving safe values.

I have attached a zipfile with the LTspice files for those playing along at home.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **November 04, 2016, 06:16:03 pm**

Jay, thank you for your effort. To be honest, I need time to get the grip on above simulations. :palm:

Sorry Jay, its just I'm not sure how above simulation translated into real circuit. :'(

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **November 07, 2016, 10:28:22 am**

BravoV and the group,

I have expanded the LTspice model that I was using earlier to make it more like a real circuit. This model should be much easier to implement in hardware:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=268342;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=268344;image)

This model uses an IXYS generated model for the MOSFET, downloaded from the IXYS website. I am not sure about the small-signal accuracy of this model with low Vds.

A large MOSFET in this application does not require large GATE currents. Here is the gate current waveform:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=268355;image)

The 100 gate resistor, isolates the op-amp from the input capacitance of the MOSFET so it is not necessary to use C-load stable op-amps.

I have attached the expanded LTspice model.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **November 08, 2016, 01:56:34 am**

Thanks a lot Jay ! :-+ Definitely this will be my nice project for the upcoming end of the year holidays.

Quote from: Jay_Diddy_B on November 07, 2016, 10:28:22 am

> This model uses an IXYS generated model for the MOSFET, downloaded from the IXYS website. I am not sure about the small-signal accuracy of this model with low Vds.

What does this mean ?

What do I need to look at, especially low Vds as in testing low voltage power supply ?

Quote from: Jay_Diddy_B on November 07, 2016, 10:28:22 am

> A large MOSFET in this application does not require large GATE currents. Here is the gate current waveform:

Is that because we don't need a really-really fast turn on/off ? Say at sub microsecond ?

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **November 08, 2016, 02:53:05 am**

Quote from: BravoV on November 08, 2016, 01:56:34 am

> Thanks a lot Jay ! :-+ Definitely this will be my nice project for the upcoming end of the year holidays.
>
> Quote from: Jay_Diddy_B on November 07, 2016, 10:28:22 am
>
> > This model uses an IXYS generated model for the MOSFET, downloaded from the IXYS website. I am not sure about the small-signal accuracy of this model with low Vds.

> What does this mean ?
>
> What do I need to look at, especially low Vds as in testing low voltage power supply ?
>
> Quote from: Jay_Diddy_B on November 07, 2016, 10:28:22 am
>
> > A large MOSFET in this application does not require large GATE currents. Here is the gate current waveform:
>
> Is that because we don't need a really-really fast turn on/off ? Say at sub microsecond ?

BravoV and the group,

This my main concern with the IXYS MOSFET model. The datasheet shows this set of curves:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=266916;image)

I don't see anything in the model that accounts for the step in Crss. The value of Crss changes by a factor of 10.

Because of this step, the MOSFET behave differently at low voltages, 3V, than it does at higher voltages say 15V. This is why I did the earlier modelling with two different input voltages.

The component values that I have given should be safe. It may be possible to get more performance out of the MOSFET, but that would require bench testing.

In a load like this, the amount you have to change the gate voltage is small. If the transconductance of the MOSFET is 5, that is a change in drain current of 5A for a 1V change in gate voltage. To increase the output current by 5A you have increase the voltage on Ciss by 1V and decrease the voltage on Crss by 1V.

If you wanted to switch this MOSFET with a 400V drain voltage, you would have to charge Ciss with 10V and discharge Crss by 400V, much bigger changes and therefore higher currents.

Here are the waveforms from the model:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=268584;image)

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **November 08, 2016, 03:30:32 am**

> Quote from: Jay_Diddy_B on November 08, 2016, 02:53:05 am
>
> > I don't see anything in the model that accounts for the step in Crss. The value of Crss changes by a factor of 10.
> >
> > Because of this step, the MOSFET behave differently at low voltages, 3V, than it does at higher voltages say 15V. This is why I did the earlier modelling with two different input voltages.
> >
> > The component values that I have given should be safe. It may be possible to get more performance out of the MOSFET, but that would require bench testing.
> >
> > In a load like this, the amount you have to change the gate voltage is small. If the transconductance of the MOSFET is 5, that is a change in drain current of 5A for a 1V change in gate voltage. To increase the output current by 5A you have increase the voltage on Ciss by 1V and decrease the voltage on Crss by 1V.
> >
> > If you wanted to switch this MOSFET with a 400V drain voltage, you would have to charge Ciss with 10V and discharge Crss by 400V, much bigger changes and therefore higher currents.

This reminds me of a famous quote ... ""The more I learn, the more I realize how much I don't know." :P

For sure once built, I will definitely report it here on low voltage range < 5 V.

Thank you Sir. :-+

Title: **Re: Dynamic Electronic Load Project**
Post by: **Floyo** on **November 10, 2016, 08:47:24 pm**

Hi Jay_Diddy_B and others,

I was wondering how the loop would change if instead of Fets Bjts were used.
The reason I'm asking is that I have recently come in the possession of a somewhat old multi
channel electronic load using the good old 2N3055 as the pass elements. The control circuit
on these however is a bit crude and non-dynamic and the single turn wire wound pots are shot, so I would like to update it a bit.
Since the beta of the 2n3055 is so low on the biggest channels (9 pass elements, 300w) there are two driver stages, one BD237 and
one 2n3055 driving the others.

I would really like to know how this added gain affects the control loop so I can design a nice new driver board.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **November 10, 2016, 08:58:48 pm**

Floyo,

BJTs should be good, may be even a little faster than the MOSFETs because you don't have to deal with the input capacitance.

The 'secret' to making a good dynamic load is to make sure that the wiring inductance between the load an the power supply is included in the analysis. I have added an RC damping circuit to help with this.

To recycle the load that you have, I would keep the power stage, any power supplies and cooling etc. and build a new control circuit.

Can you draw the schematic of the power stage?

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Floyo** on **November 10, 2016, 10:11:38 pm**

Here is the power stage as I just traced it out, the full schematic I did some time earlier. They might contain some errors, but it seems reasonable. This is of the biggest stages, where the power stage is basically a darlington driven by the bd237, the smaller stages are driven straight by the bd237.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **November 10, 2016, 10:18:47 pm**

Do you know the value of the shunt?

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **November 11, 2016, 03:15:47 am**

Quote from: Floyo on November 10, 2016, 08:47:24 pm

> Hi Jay_Diddy_B and others,
>
> I was wondering how the loop would change if instead of Fets Bjts were used.
> The reason I'm asking is that I have recently come in the possession of a somewhat old multi
> channel electronic load using the good old 2N3055 as the pass elements. The control circuit
> on these however is a bit crude and non-dynamic and the single turn wire wound pots are shot, so I would like to update it a bit.
> Since the beta of the 2n3055 is so low on the biggest channels (9 pass elements, 300w) there are two driver stages, one BD237 and
> one 2n3055 driving the others.
>
> I would really like to know how this added gain affects the control loop so I can design a nice new driver board.

Based on the information in the later posts I have put a model together for a load based on the BJT power stage.

LTspice Model

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=269375;image)

The **M=8** in SPICE means 8 devices in parallel. This saves having to draw all the parallel parts.

Plot V(a)/V(b) to plot the loop gain.

Modelling Results

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=269377;image)

I have picked component values for the op-amp stage to stabilize the control loop. I also guessed the value of the emitter resistors and the shunt.

I have attached the LTspice model if you want to try other values.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **November 11, 2016, 03:30:03 am**

Hi,

This last model didn't look right. I played around a bit and I found that the model for Q3 is probably wrong. If I substitute a different transistor in the Q3 location I get much better results.

I was able to reduce the value of C1 and get more bandwidth.

Modified Model

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=269381;image)

Results

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=269383;image)

The performance is the same as the MOSFET load presented earlier.

Remember to check the Safe Operating Area, FBSOA, on the BJT datasheet.

I have attached the model.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **November 11, 2016, 03:41:55 am**

Hi group,

With the 2SCR574D transistor in the Q3 position. Here is the modeling for the time domain.

Model

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=269386;image)

Results

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=269388;image)

I have attached the model.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **BravoV** on **November 11, 2016, 04:58:26 am**

Jay,

Is it ok to replace the pulse generator part with this highlighted ones using simple 555 ics ? without disrupting the control loop ?

I just love the extra adjustabilities like the freq and pulse width. :P

Its from TI AN-1733 (http://www.ti.com/lit/an/snoa507/snoa507.pdf).

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=269401;image)

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **November 11, 2016, 05:10:14 am**

Quote from: BravoV on November 11, 2016, 04:58:26 am

> Jay,
>
> Is it ok to replace the pulse generator part with this highlighted ones using simple 555 ics ? without disrupting the control loop ?
>
> I just love the extra adjustabilities like the freq and pulse width. :P

Yes, you can use the 2x 555 circuit. You may have to play with C2 and C4 to the frequency range and pulse width that you desire.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Floyo** on **November 11, 2016, 07:59:10 am**

Hi Jay_Diddy_B,

Thanks for the simulations, that gives me enough info to start
prototyping. The values of the shunt vary quite a bit between the four channels since their ratings are quite different.
The data can be found attached, for completeness sake. The voltage ratings on the channels seem to be chosen so that the rated current produces the rated power dissipation
at that voltage. Keeping in mind the SOA curves they will handle voltages quite a bit higher. I might end up replacing the transistors of some of the smaller channels with some higher VCEmax types, with matching soa to make some higher voltage channels.

I'll have to see if the 1mOhm shunt of the 60A channel gives satisfactory performance at the low end of the scale, if not
I might replace it with something more reasonable for a 20A range or so.

The rest of the load is fine to reuse, it just needs some minor mods to make it become a rather handy tool. The fans in particular will receive some "dimming", 2*20w of fan is *Loud*.

Title: **Re: Dynamic Electronic Load Project**
Post by: **CK345** on **December 04, 2016, 06:47:22 am**

Analysis of a great :-+

Title: **Re: Dynamic Electronic Load Project**
Post by: **CK345** on **December 08, 2016, 11:36:05 am**

Hello:
why this figure gain margin is negative, the loop is stable?According to the loop analysis gain abundant negative loop unstable!

Title: **Re: Dynamic Electronic Load Project**
Post by: **henken** on **January 31, 2017, 01:02:37 pm**

Hello,

Very informative analysis going on here. Thanks a lot for making it all available for us to see and learn.

I have been playing around with variations of the first time domain model in this thread, and I have a few questions from the perspective of a beginner in the subject.

I am looking into different variations of supply voltages, MOSFETS, number of MOSFET stages, and current limits. The main variables affecting the current drawn from the source are (as you of course know): the reference voltage (vref), the resistance of the current sense resistors, and the number of MOSFET stages. I understand the relationships between these, how they affect the current draw, and how they put constraints on the supply rails.

My questions are all related to which of these things can be changed without significantly affecting the stability analysis (I am still not understanding the subject well enough to trust my own analysis).

How sensitive is the system with regards to changes in:

1. Number of MOSFET stages. My uneducated guess is that this is not an important factor since they are all individual control loops.

2. Current sense resistor value. Again my guess here is that I can vary this within reasonable limits to adjust the current draw.

3. The reference voltage. My guess here is that this can also be changed somewhat without seriously affecting stability. For example, can a 1.25V reference be used instead of 2.5V? This would matter if a -5 rail is available instead of a -9V rail.

4. MOSFET choice. This is of course a major factor in the stability analysis. I can't source the IRFZ20 MOSFETs for a reasonable price, so I am looking for alternatives that don't require a completely new analysis. The MOSFET model used in the initial simulation model is the IRF530. I assume that this model will work instead of the IRFZ20.

In short:
I am thinking of implementing this circuit with 4xIRF530 to spread the heat over more devices. Thus, I would need to either change the reference voltage to 1.25V, or increase the current sense resistor to 0.2? in order to keep the same 5A ranges. My hope is that this won't negatively affect the stability analysis already made.

Thanks in advance for any further advice.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **February 01, 2017, 02:01:49 am**

Quote from: henken on January 31, 2017, 01:02:37 pm

Hello,

Very informative analysis going on here. Thanks a lot for making it all available for us to see and learn.

I have been playing around with variations of the first time domain model in this thread, and I have a few questions from the perspective of a beginner in the subject.

I am looking into different variations of supply voltages, MOSFETS, number of MOSFET stages, and current limits. The main variables affecting the current drawn from the source are (as you of course know): the reference voltage (vref), the resistance of the current sense resistors, and the number of MOSFET stages. I understand the relationships between these, how they affect the current draw, and how they put constraints on the supply rails.

My questions are all related to which of these things can be changed without significantly affecting the stability analysis (I am still not understanding the subject well enough to trust my own analysis).

How sensitive is the system with regards to changes in:

1. Number of MOSFET stages. My uneducated guess is that this is not an important factor since they are all individual control loops.

2. Current sense resistor value. Again my guess here is that I can vary this within reasonable limits to adjust the current draw.

3. The reference voltage. My guess here is that this can also be changed somewhat without seriously affecting stability. For example, can a 1.25V reference be used instead of 2.5V? This would matter if a -5 rail is available instead of a -9V rail.

4. MOSFET choice. This is of course a major factor in the stability analysis. I can't source the IRFZ20 MOSFETs for a reasonable price, so I am looking for alternatives that don't require a completely new analysis. The MOSFET model used in the initial simulation model is the IRF530. I assume that this model will work instead of the IRFZ20.

In short:
I am thinking of implementing this circuit with 4xIRF530 to spread the heat over more devices. Thus, I would need to either change the reference voltage to 1.25V, or increase the current sense resistor to 0.2? in order to keep the same 5A ranges. My hope is that this won't negatively affect the stability analysis already made.

Thanks in advance for any further advice.

Thank you for your kind words.

Let me answer your questions in the same order:

1) You are correct, each MOSFET has its own control loop, so the number of stages used in parallel does not impact the control loop stability.

2) If you examine the voltage gain of the MOSFET stage from the gate of the MOSFET to the source of the MOSFET as shown in this LTspice model:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=288860;image)

The results are:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=288862;image)

So the maximum gain is 0dB if the Source resistance is high and the transconductance of the MOSFET is high. The gain is reduced if the transconductance is lowered or the source resistance is lowered.

3) The reference voltage does not directly impact the gain.

4) Choice of MOSFET

The Vishay datasheets show that the IRFZ20 and the IRF530 are similar. The gm for both MOSFETs changes with the operating point. They do not

show gm directly but a graph like this:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=288864;image)

The transconductance is the change in Id versus the change in Vgs.

These MOSFETs are close enough to be considered equivalent.

When the control loops are designed there should be a least 6dB of gain margin. 6dB is factor of 2. so this allows for some significant changes while maintain stability.

Do not forget the RC damping network. This is critical in making a good load.

Good luck with your project, please share your results.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **henken** on **February 01, 2017, 09:38:50 pm**

Thanks for your response!

The reference voltage and MOSFET answers are clearly understood, and make great sense (just turning the potentiometers will affect the set voltage anyhow). The source resistor answer is a bit more difficult to understand.

Quote from: Jay_Diddy_B on February 01, 2017, 02:01:49 am

> 2) If you examine the voltage gain of the MOSFET stage from the gate of the MOSFET to the source of the MOSFET as shown in this LTspice model:
>
> [pic]
>
> The results are:
>
> [pic]
>
> So the maximum gain is 0dB if the Source resistance is high and the transconductance of the MOSFET is high. The gain is reduced if the transconductance is lowered or the source resistance is lowered.
>
> ...
>
> When the control loops are designed there should be a least 6dB of gain margin. 6dB is factor of 2. so this allows for some significant changes while maintain stability.

Clearly the gain increases when the resistance is increased. I'm not entirely clear how much this affects the actual stability in the end, and how it ties together with the actual gain margin. I need to read more about this subject. I will keep the same 0.1 milliohm source resistance anyway, and choose a different reference.

Quote from: Jay_Diddy_B on February 01, 2017, 02:01:49 am

> Do not forget the RC damping network. This is critical in making a good load.

I picked up on that from your earlier posts. :-+

After some further testing with the time domain model, I found that the current slew rate seems severely limited. I actually don't have a clear idea about how large current slew rate is actually needed for some basic transient testing, but I looked around and found some application notes on the subject:

http://cds.linear.com/docs/en/application-note/an104f.pdf (http://cds.linear.com/docs/en/application-note/an104f.pdf)
http://www.ti.com/lit/an/snoa507/snoa507.pdf (http://www.ti.com/lit/an/snoa507/snoa507.pdf)

They mention current slew rates at least in the tens of A/us (TI mentions 50A/us, and the rise times in LT's figures are much faster than what I can achieve in the simulation).

One limitation in the model seems to be the LT1013 op-amp, which only has a slew rate of 0.4 V/us. I tried changing the op-amps to ideal ones with the same slew rate as the TL074. This gave me much faster edges on the stages before the actual MOSFET driver. Doing that didn't speed up the final current rise times significantly, which seems to be due to the 220pF capacitors at the output stage.

I tried to measure the slope of the drawn current in LTspice. The current seems to peak at less than 1 A/us in the model you provided. Removing the 220p caps yielded a peak slope of 45 A/us, when the op-amps were interchanged with 13 V/us op-amps. Of course the lead inductance plays into all this, and I had to reduce it to get these figures.

So, I have a couple more questions:

1. Would it be desirable to have faster rise times, or are they already fast enough for basic load transient testing?

2. How could the circuit be modified to get faster rise times, while maintaining stability?

Quote from: Jay_Diddy_B on February 01, 2017, 02:01:49 am

> Good luck with your project, please share your results.

Thanks. it's going to take a while until I have anything to show. I must finalize the design, source parts, and get enough free time to put it together.

Cheers

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **February 03, 2017, 11:55:29 pm**

Quote from: henken on February 01, 2017, 09:38:50 pm

> Thanks for your response!
>
> snip ...
>
> So, I have a couple more questions:
>
> 1. Would it be desirable to have faster rise times, or are they already fast enough for basic load transient testing?
>
> 2. How could the circuit be modified to get faster rise times, while maintaining stability?

There are some faster loads out there. I have done a quick survey:

HP 6060A or 6060B:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=289730;image)

The HP6060B has two ranges 0-30A high range and 0-3A on the low range. The slew rate is adjustable.

The BK Precision manual shows:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=289732;image)
(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=289734;image)
(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=289736;image)

So the BK is not as fast as the HP/Agilent.

How fast is good enough?

We can explore this in LTspice. Here is a model with a dynamic load. The transition time is being stepped between 1us to 50us in a 1-2-5 sequence.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=289740;image)

Here are the results:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=289742;image)

You can see that the transition time has little effect until the transition time is similar to the response time of the power supply.

V=Ldi/dt constraints

Inductance between the load and the power supply will constrain how fast the load current can be stepped.

Jim Williams in Linear Technology AN133
Link: http://cds.linear.com/docs/en/application-note/an133f.pdf (http://cds.linear.com/docs/en/application-note/an133f.pdf)

describes a very fast load, suitable for testing low voltage high current power supplies. But, you can see how just 20nH of inductance wrecks the performance:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=289738;image)

This is a special load for testing core supplies for FPGAs. It would not be a very good general purpose load.

You have the right idea, small MOSFETs, faster op-amps. But you have to follow the procedure outlined in this thread.

I do not want a load that is marginally stable when testing power supplies.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **henken** on **February 04, 2017, 01:38:29 am**

Thanks for the reply, and for the effort you put into them. Much appreciated.

I also found AN133. Good read.

Now I am diving into learning more about AC analysis (there's a huge amount to learn here :)). Meanwhile, I have been playing around with the AC analysis in LTspice, using your model as a starting point.

By random testing, I think I have found a configuration with much higher bandwidth that appears to be stable (from my unexperienced read of the Bode plots). This is probably all obvious to someone in the know, but I'll put it out anyhow.

1. Use a higher bandwidth op-amp
2. Reduce feedback capacitor (what is the proper name here?) from 220pF to 20pF.
3. Remove, or reduce the base resistor to the MOSFET (Downsides?)
4. The changes above cause instability -> Change the compensation by removing the resistor, and reducing the compensation capacitor to 1uF. (This makes the configuration look just like a regular bypass capacitor, which does fill the same role. I am not sure if that's improper in some way in the context of an electronic load. Seems fine.)

With these changes, I can get very clean waveforms, with a bandwidth of ~500kHz, and ~7A/us peak (4 source MOSFETS), even with plenty of lead inductance. Undoubtedly, things may turn out differently when I build this thing. The nice thing is that the architecture seems general enough to allow lots of experimentation with different component values, so if things turn out unstable, there is no need to start over with a new board (as long as the layout is good enough).

One issue I found with the circuit in this thread, is that the square wave reference voltage for pulsed loads has large spikes. The spikes are not present at the oscillator itself, but only after the transistor switching the reference voltage. Some kind of parasitic of the transistor must be to

blame here. The spikes cause over/undershoot at the sourced current, which go away when clean pulses are used. Changing transistor has an effect, but I have not been able to get rid of the spikes that way. Ideas?

More questions:

1. What is the point of manually setting up the frequency stepping, and accessing the Bode plots through the error log vs. using a regular .ac command for sweeping the frequency?
2. Any advice on cleaning up the input pulses without overly impacting the rise-times?

I am of course looking into these things myself as well, but any pointers are appreciated.

Best regards

Edit: Ehm. Too quick to celebrate perhaps. Measuring I(R5) as you do, there is ringing with my modified configuration. The current waveform through the load is clean though. I'm not sure how to approach this from a power supply test perspective. Need to do more research..

Title: **Re: Dynamic Electronic Load Project**
Post by: **Aroogz** on **July 31, 2017, 12:06:49 pm**

Hi ,

Nice and Interesting work Jay_Diddy_B. well Done.

However, I am trying to build a dynamic electronic load with the following specs:

- frequency: 0.1Hz - 10kHz
- Voltage range: 0 - 40V
- current range of 0 - 30A

Please what's your advice and could this be modified to accommodate the requirements?

Thankyou.

Aroogz.

Title: **Re: Dynamic Electronic Load Project**
Post by: **fcb** on **July 31, 2017, 01:45:40 pm**

Quite ambitious.

Probably start with building a 1A 40v load and getting that working at 10KHz. Then build up from there.

Also, what is your 10KHz - a squarewave? sine? will make quite a difference to your acceptance criteria.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **July 31, 2017, 03:56:14 pm**

Quote from: Aroogz on July 31, 2017, 12:06:49 pm

> Hi ,
>
> Nice and Interesting work Jay_Diddy_B. well Done.
>
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>
> - frequency: 0.1Hz - 10kHz
> - Voltage range: 0 - 40V
> - current range of 0 - 30A
>
> Please what's your advice and could this be modified to accommodate the requirements?
>
> Thankyou.
>
> Aroogz.

Thank you for the kind words.

You indicate that you need 0-40V and 0-30A, do you need 1200W?

Because the MOSFETs are being used in the Linear region, 40W per MOSFET is a sensible target. The HP 6060A/B uses 8 MOSFETs for 300W.

Each MOSFET needs it own op-amp to control the current.

Quote from: fcb on July 31, 2017, 01:45:40 pm

> Quite ambitious.
>
> Probably start with building a 1A 40v load and getting that working at 10KHz. Then build up from there.
>
> Also, what is your 10KHz - a squarewave? sine? will make quite a difference to your acceptance criteria.

This is great advice!!

My original circuits has two MOSFETs, but it can be extended to many.

The oscillator in the circuit shown is set to run at 330Hz. This is a good frequency for testing most power supplies. There is no reason why it can't be set higher.
To get high bandwidth in the load you need to use small MOSFETs and lots of them.

Good luck !!

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Aroogz** on **August 02, 2017, 10:19:47 am**

Quote from: fcb on July 31, 2017, 01:45:40 pm

> Quite ambitious.
>
> Probably start with building a 1A 40v load and getting that working at 10KHz. Then build up from there.
>
> Also, what is your 10KHz - a squarewave? sine? will make quite a difference to your acceptance criteria.

Thanks very much for your reply.

The 10kHz is a sine wave signal (of about 250mAp-p) which will ride ontop of a set DC current of the load.

Regards,
Aroogz.
The goal is to use this to perform a frequency sweep (0.1Hz - 10KHz) at set DC load point.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Aroogz** on **August 02, 2017, 10:36:41 am**

Quote from: Jay_Diddy_B on July 31, 2017, 03:56:14 pm

> Quote from: Aroogz on July 31, 2017, 12:06:49 pm
>
> > Hi ,
> >
> > Nice and Interesting work Jay_Diddy_B. well Done.
> >
> > However, I am trying to build a dynamic electronic load with the following specs:
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> > - Voltage range: 0 - 40V
> > - current range of 0 - 30A
> >
> > Please what's your advice and could this be modified to accommodate the requirements?
> >
> > Thankyou.
> >
> > Aroogz.
>
> Thank you for the kind words.
>
> You indicate that you need 0-40V and 0-30A, do you need 1200W?
>
> Because the MOSFETs are being used in the Linear region, 40W per MOSFET is a sensible target. The HP 6060A/B uses 8 MOSFETs for 300W.
>
> Each MOSFET needs it own op-amp to control the current.
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> Quote from: fcb on July 31, 2017, 01:45:40 pm
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> To get high bandwidth in the load you need to use small MOSFETs and lots of them.
>
> Good luck !!
>
> Regards,
>
> Jay_Diddy_B

Jay_Diddy_B,

Thank for your advice.

The requirement is just about 200W.

Regards,
Aroogz.

Title: **Re: Dynamic Electronic Load Project**
Post by: **fcb** on **August 02, 2017, 10:55:24 am**

Quote from: Aroogz on August 02, 2017, 10:19:47 am

> Quote from: fcb on July 31, 2017, 01:45:40 pm
>
> > Quite ambitious.
> >
> > Probably start with building a 1A 40v load and getting that working at 10KHz. Then build up from there.
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> Thanks very much for your reply.

> The 10kHz is a sine wave signal (of about 250mVp-p) which will ride ontop of a set DC current of the load.
>
> Regards,
> Aroogz.
> The goal is to use this to perform a frequency sweep (0.1Hz - 10KHz) at set DC load point.

250mVpk-pk.. this is a DC load - that doesn't make sense as an 'output' of the load.

Do you mean that you will adjust the load till you get 250mVpk-pk ripple?

Title: **Re: Dynamic Electronic Load Project**
Post by: **Rbastler** on **August 02, 2017, 10:57:27 am**

> Quote from: fcb on August 02, 2017, 10:55:24 am
>
> > Quote from: Aroogz on August 02, 2017, 10:19:47 am
> >
> > > Quote from: fcb on July 31, 2017, 01:45:40 pm
> > >
> > > > Quite ambitious.
> > > >
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> > > >
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> > >
> > > Thanks very much for your reply.
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> > > The 10kHz is a sine wave signal (of about 250mVp-p) which will ride ontop of a set DC current of the load.
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> > > Regards,
> > > Aroogz.
> > > The goal is to use this to perform a frequency sweep (0.1Hz - 10KHz) at set DC load point.
> >
> > 250mVpk-pk.. this is a DC load - that doesn't make sense as an 'output' of the load.
> >
> > Do you mean that you will adjust the load till you get 250mVpk-pk ripple?

Maybe he wants a pulsed load ?

Sent from my A0001 using Tapatalk

Title: **Re: Dynamic Electronic Load Project**
Post by: **Aroogz** on **August 02, 2017, 11:17:08 am**

> Quote from: fcb on August 02, 2017, 10:55:24 am
>
> > Quote from: Aroogz on August 02, 2017, 10:19:47 am
> >
> > > Quote from: fcb on July 31, 2017, 01:45:40 pm
> > >
> > > > Quite ambitious.
> > > >
> > > > Probably start with building a 1A 40v load and getting that working at 10KHz. Then build up from there.
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> > > The 10kHz is a sine wave signal (of about 250mVp-p) which will ride ontop of a set DC current of the load.
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> >
> > 250mVpk-pk.. this is a DC load - that doesn't make sense as an 'output' of the load.
> >
> > Do you mean that you will adjust the load till you get 250mVpk-pk ripple?

Yes you are right....it's a DC load with the sinusoid (ripple on top). So this could be an analog signal that modulates the load.

It's sort of a transient load with a dc offset. For instance, I have a load of 1A with a sinusoid (to be swept over the frequency range) of amplitude of about 100mA. This will be the current drawn from the supply.

I am sorry if this isn't clear, but I will be willing to make any necessary further clarifications.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Aroogz** on **August 02, 2017, 11:19:46 am**

> Quote from: fcb on August 02, 2017, 10:55:24 am
>
> > Quote from: Aroogz on August 02, 2017, 10:19:47 am
> >
> > > Quote from: fcb on July 31, 2017, 01:45:40 pm
> > >
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> >
> > 250mVpk-pk.. this is a DC load - that doesn't make sense as an 'output' of the load.
> >
> > Do you mean that you will adjust the load till you get 250mVpk-pk ripple?

I am sorry, I meant to say 250mA.

Title: **Re: Dynamic Electronic Load Project**
Post by: **edzyj** on **August 21, 2017, 07:17:05 am**

Nice and Interesting work ! :-+

Title: **Re: Dynamic Electronic Load Project**
Post by: **capt bullshot** on **May 29, 2018, 07:01:47 am**

Great job, Jay_Diddy_B!

Title: **Re: Dynamic Electronic Load Project**
Post by: **dardosordi** on **August 13, 2018, 12:36:23 pm**

Hi,

first I want to thank Jay_Diddy_B for your effort on this project, Im looking into building a 2nd load since my first DIY one melted its plastic case... :-BROKE :palm:

Im considering ways to get away with a single supply. I see why you did this as a non-inverting amp in order to easily add up the voltages, but it makes you to use symmetric supply.

Could buffering the oscillator help with this and keep it single supply ?

Im using this crude simulator since I have no idea where to start with LT Spice (definitively on my list of to-do things). Using a BJT since I couldn't get the mosfet won't behave linear in it.

Here's the rather big url:

https://www.falstad.com/circuit/circuitjs.html?... (https://www.falstad.com/circuit/circuitjs.html?cct=$+1+0.000005+10.20027730826997+50+5+43%0Ag+592+384+592+400+0%0Av+656+112+656+64+0+0+40+50+0+0+0.5%0Ar+592+272+592+320+(

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 14, 2018, 01:50:38 pm**

Quote from: dardosordi on August 13, 2018, 12:36:23 pm

> Hi,
>
> first I want to thank Jay_Diddy_B for your effort on this project, Im looking into building a 2nd load since my first DIY one melted its plastic case... :-BROKE :palm:
>
> Im considering ways to get away with a single supply. I see why you did this as a non-inverting amp in order to easily add up the voltages, but it makes you to use symmetric supply.
>
> Could buffering the oscillator help with this and keep it single supply ?
>
> Snip ..

Hi,

Thank you for your kind words.
Yes, the inverting configuration was used to make the summing easier. It also allows a wide selection of op-amps because they don't need to be rail-to-rail input and output.

Having said that here is an idea for you:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=497537;image)

*** There is a mistake in the formula - ~~I will fix later~~ ***

Corrected Version:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=499355;image)

The circuit combines the pulsating signal, Transient, with the non-inverting signal Vref.

The whole circuit becomes:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=497543;image)

I have not built this circuit, so I do not know how well it will work.

I have attached the LTspice models.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **dardosordi** on **August 17, 2018, 02:31:29 am**

Thank you very much! I will build it on the weekend. Is LM324 fine for this? I also have TL082/072.

Thanks again!

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 17, 2018, 03:00:32 am**

> Quote from: dardosordi on August 17, 2018, 02:31:29 am
> Thank you very much! I will build it on the weekend. Is LM324 fine for this? I also have TL082/072.
>
> Thanks again!

Dardosordi,

Use the LM324 op-amps. The LM324 common range includes the negative rail. this essential in the single supply version. The TL072/82 common mode range does not include the negative rail, so these will not work with a single supply.

With the LM324 you may need to reduce the bandwidth replace C1 and C3 with 1nF or 2.2nF to prevent oscillation.

Let us know how it works.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 19, 2018, 05:50:14 am**

Hi, JDB, Thanks for referring me to this load project. Your work, on this, is wonderful, impressive and gracious.

You are helping me, on another thread, to troubleshoot a load. But, its design is not your favorite. So, I am considering your design, to replace it, once I have used the present fault, as a learning opportunity.

Firstly, I looked up the difference between a Constant Current and a Dynamic load. Okay, I get it... The dynamic load allows the current draw to be varied (at a known frequency), so that the characteristics of the supply can be observed.

Here is my first Noob question. Can this circuit be modified, to turn off the oscillation? In other words, have a Constant Current/Dynamic Current option? If it is not too much to ask, what would be that design? Is it as easy as adding a switch, to by pass the oscillator portion, of the circuit?

Thanks!

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 19, 2018, 02:33:31 pm**

> Quote from: t1d on August 19, 2018, 05:50:14 am
> Hi, JDB, Thanks for referring me to this load project. Your work, on this, is wonderful, impressive and gracious.
>
> You are helping me, on another thread, to troubleshoot a load. But, its design is not your favorite. So, I am considering your design, to replace it, once I have used the present fault, as a learning opportunity.
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> Firstly, I looked up the difference between a Constant Current and a Dynamic load. Okay, I get it... The dynamic load allows the current draw to be varied (at a known frequency), so that the characteristics of the supply can be observed.
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> Here is my first Noob question. Can this circuit be modified, to turn off the oscillation? In other words, have a Constant Current/Dynamic Current option? If it is not too much to ask, what would be that design? Is it as easy as adding a switch, to by pass the oscillator portion, of the circuit?
>
> Thanks!

T1D,

There are two controls, one sets the constant level, the other sets the pulsed level. To get constant current, you simply turn the pulsed control fully counter clockwise:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=63299;image)

Using a pulsed load is used to look how a power supply responds to step changes in current.

It would be like test driving a car, without going round corners, if you tested a power supply without stepping the load current.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 19, 2018, 09:16:07 pm**

> Quote from: Jay_Diddy_B on August 19, 2018, 02:33:31 pm
> There are two controls, one sets the constant level, the other sets the pulsed level. To get constant current, you simply turn the pulsed control fully counter clockwise:

Doh! Too simple...

> Quote from: Jay_Diddy_B on August 19, 2018, 02:33:31 pm
> Using a pulsed load is used to look how a power supply responds to step changes in current.

Yes, I am understanding, that part...Though I may not have said that, clearly.

As always, thank you very much.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 20, 2018, 07:25:27 pm**

Hi group,

One question that comes up a lot in the discussion of electronic loads is:

Why is one op-amp per MOSFET better?

Most commercial loads, like the HP6060A/B and 6050x use one op-amp MOSFET.

The MOSFET datasheet has the answer:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=502631;image)

This graph is from the obsolete On-Semi MTP3055V Datasheet. Other parts have similar characteristics.
If you fix the Vgs at 4V you will see that the drain current, Id, has a positive temperature coefficient. As the device gets hotter, there is more Id and the temperature rises. We have the conditions for thermal runaway.

Other hand if we have one op-amp MOSFET each of the drain currents is controlled separately and there is no possibility of thermal runaway.

Modelling

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=502637;image)

Here I have a load with two identical MOSFETs. There is no issue, the Drain currents are perfectly balanced.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=502643;image)

If I unbalance the circuit by placing a 0.1V source in series with the MOSFET gate. This is the same as the MOSFET having a different transfer characteristic:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=502649;image)

The currents are no longer balanced:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=502655;image)

Separate Source resistors

One technique that is often employed is to use separate source resistors for each MOSFET. We can modify the load to this:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=502661;image)

R7 and R8 were added so that the load regulates the average of the current in M4 and M5.

Results of separate source resistors.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=502667;image)

Although there is an improvement, it is only slight. By inspection, for this to work well the voltage across R3 and R6 must be 'large' compared to differences in Vgs transfer characteristics. This means high power dissipation in these resistors.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 20, 2018, 10:59:51 pm**

Thermal runaway... Brilliant! Thanks, Jay. Excellent explanation, too. I suggest also posting this explanation, as its own thread, to make it easier to find the information.

Speaking of the MTP3055V, I have a supply of the MTP3055VL and AD8032ANZ Dual Op Amp. Would these work, for your single supply design?

I am building the single supply design schematic and PCB, in Kicad, in order to have the PCB manufactured. I will be glad to contribute the files, once they are finished. They will greatly reduce the work needed, to make personal customizations.

I will probably need help, with proofing the schematic and laying out the PCB (Though I am not bad, at PCB Layout. But, more eyes are better.) And, with multiplying the design... I am going for ~100 watts, to test PSUs, having 30v and 3a.

One issue, that I already see, is that the Op Amp schematic symbol dictates the PCB footprint pin assignments. So, the components will need to be placed, on the PCB, and, then, the Op Amp schematic symbols will need to be swapped around, to facilitate the best placement of the components, on the PCB.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 21, 2018, 12:25:05 am**

Jay, as I am not a professional EE, I am not familiar, with this symbolism. I take it that the circles indicate complete nodes(correct term?) The one on the left being the 9 volt regulator, with all its supporting components, and the one on the right being the 2.5 volt reference, with all its supporting components? Correct?

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 21, 2018, 03:08:34 am**

Quote from: t1d on August 21, 2018, 12:25:05 am

> Jay, as I am not a professional EE, I am not familiar, with this symbolism. I take it that the circles indicate complete nodes(correct term?) The one on the left being the 9 volt regulator, with all its supporting components, and the one on the right being the 2.5 volt reference, with all its supporting components? Correct?

These are symbols used in SPICE schematics. The circle with a V designation is a voltage source.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=502649;image)

In this schematic:

V1 is a 9V power source, it could be a regulated power supply or simply a 9V battery.

V2 is the power supply being tested

V3 is a voltage source used to generate an offset.

In the other schematic there is a 2.5V voltage source, this could be a voltage output reference (example LTC6655CHMS8-2.5) or a shunt regulator(example TL431) and its associated components.

It is a shortform, used in modelling.

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 21, 2018, 04:57:24 am**

Background: KiCad components need to be shown, on the schematic, in order for the traces to be tied, pin-to-pin, on the board. This requirement dictates clarity, in the following matters. Meaning, I am not trying to be picky.

V1 is a 9V power source, it could be a regulated power supply or simply a 9V battery.
Here, I have added a L7809, with supporting caps.

V2 is the power supply being tested
V3 is a voltage source used to generate an offset.
V2 is not included, on the Single Supply schematic. I take it that, on the Single Supply schematic, R5, L1 and V3 should be removed and a DUT input connector (+ and -) should be added, in their place. Correct? Yes, I see that, on the "Real Schematic," now. Added same.

In the other schematic there is a 2.5V voltage source, this could be a voltage output reference (example LTC6655CHMS8-2.5) or a shunt regulator(example TL431) and its associated components.
Here, I have added the LT6656 voltage reference, with supporting caps.

I added the meter output terminating resistor, R19/51R, and its connector.

Please clarify the suffix, of the values, of these resistors. I could guess, but that is dangerous. Such as R19 is a 51R resistor…

R1 100
R2 0.1
R20 100
R21 0.1
R23 3300
R4 3300
R5 0.1
R9 2.2

I am nearly finished. Do you use KiCad? I will provide a jpg, too, as the schematic will, now, be ready to use, without need of researching the items, above.

I am really enjoying this adventure. Thanks!

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 23, 2018, 03:22:38 am**

It is my goal to create a schematic, in KiCad, of the Single Supply Dynamic Load Circuit, to share and to use to produce a working board, for a finished device. Attached is my Rough Draft, of same, in pdf format. I will provide the KiCad files, after everything is finalized.

This schematic is a combination of the Single Supply Model and Jay's Real Schematic. I have medical difficulties that make it nearly impossible to transpose visual information… (Flip a perf board over and solder it up… Forget-about-it…) So, I am not confident of my work.

Additional Questions, to my post #80:
#1) Red Block #1: The two pots (R21 and R22) and the three resistor divider network (R12, R13 and R14) were added, to the schematic, from JRS. I need someone to a) confirm the placements, within the circuit, and b) the values of the pots and resistors.
#2) Red Block #2: It is my guess that this resistor divider network is only for the purpose of modeling the circuit and should be removed, from the actual schematic. Correct?
#3) Some items, such as the voltage regulator, are substituted, for modeled items, from known, common applications.
#4) I need lots of folks, to proof the entire schematic, for errors, missing items, etc. The more eyes, the better.
#5) I would also like to know, if I can substitute AD8032ANZ's, for the Op Amps, and MTP3055VL's, for the MOSFETs. I plan to use five banks, to create ~100 watt capability.

I appreciate your help and look forward to finishing this up, with you.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 23, 2018, 09:58:29 pm**

T1D and the group,

The circuit presented by T1d has some issues:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=504965;image)

The op-amp U1B in configured as an inverting amplifier. It has positive inputs and a single, positive, supply. The output will try and go negative.

Because there is only a positive supply, the output of U1B will always be 0V.

Remember, I have not built the single-supply version.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 23, 2018, 10:22:44 pm**

Hi group,

Here is a little more modelling on the single-supply circuit that I proposed. I have modified the potentiometers to be voltage controlled. The control voltage is 0-100V 0V is fully ccw and 100V is fully cw.
These are representing normal, linear 10K, potentiometers, like those used when I built the dual supply version.

This is the new model:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=504971;image)

The results are:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=504977;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=504983;image)

I have attached the model for those playing along at home.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 24, 2018, 12:33:23 am**

Quote from: Jay_Diddy_B on August 23, 2018, 09:58:29 pm

> The circuit presented by T1d has some issues:

Jay, please know that I was not intending to propose any engineering, of your circuit... I was just trying to turn the Single Supply model into a schematic. I was attempting to combine the SS model, with the pots, from the dual model. I really made a mess, of that. It makes perfect sense, now that you have labeled the pots.

The new pfd, of the Revised Single Supply schematic, is attached. Please proof it. If it is okay, I will post the KiCad files.

I left out some of the narrative text, that I thought was for modeling purposes. If that text should be on the schematic, please let me know.

Would it be okay to bread board the circuit? Or, will there be issues... Voltage, amperage, capacitance? I would think that I would not want to run it above 1v/1a. If I can Bbd it, I will order the exact parts, from Mouser.

Thanks, for all that you are doing, for us.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 24, 2018, 01:22:46 am**

At the risk of adding project creep, I have two questions:
1) If I wanted to add a switch, to bring in, or cut out, the oscillation section, without spinning the knob, where would it go?
2) If using an independent function generator would add functionality greater than the on-board oscillator, where, in the circuit, should it be injected? Just use the type jack that breaks the circuit? What issues might there be?

Non-creep question:
Would there be any great advantage to using ten-turn pots?

Thanks!

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 24, 2018, 04:41:43 am**

Hi T1D and the group,

It is so much easier to do this with a negative rail. I would go back to this circuit, from the first post in this thread:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=59885;image)

I would sum the external signal into U1B by adding another 100K resistor to pin 6, the summing node.

I would convert it to single supply operation by adding a negative voltage generator. Here is an idea based on a 555 timer:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=505217;image)

And the output of the circuit:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=505223;image)

I have attached the model.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 25, 2018, 06:11:09 am**

I have finished the rough draft, of the Dual Supply model. Here is the pfd.

I added some options, for the Vref, the V Supply and the number of MOSFETs, based on the parts I have in my stocks and what I intend to build. Take your pick and delete the remainder. Or, add your own twist.

I will post the KiCad files, as soon as I clean up a few details. To do that, I could use some help, with these (bumped) questions:
Please clarify the suffix, of the values, of these resistors. I could guess, but that is dangerous. Such as R19 is a 51R resistor... These component reference numbers are for the Single Supply model, that I posted above, but they need to be updated in the Dual Model, too.

R1 100
R2 0.1
R20 100
R21 0.1
R23 3300
R4 3300
R5 0.1
R9 2.2

Would it be okay, to bread board the circuit? Or, will there be issues... Voltage, amperage, capacitance? I would think that I would not want to run it above 1v/1a. If I can Bbd it, I will order the exact parts, from Mouser.

At the risk of adding project creep, I have two questions:
1) If I wanted to add a switch, to bring in, or cut out, the oscillation section, without spinning the knob, where would it go?
2) If using an independent function generator would add functionality greater than the on-board oscillator, where, in the circuit, should it be injected? Just use the type jack that breaks the circuit? What issues might there be?
3) If two dual op amps are used, I would think that any unused units might be used to create a sign wave, too, in the oscillator section.

Non-creep question:
Would there be any great advantage to using ten-turn pots?

I still need lots of help with proofing, on both models. Please, please, oh please and thank you. <grin>

Title: **Re: Dynamic Electronic Load Project**
Post by: **Wolfgang** on **August 25, 2018, 01:00:04 pm**

Hi,

you probably know that there are specialized ICs to create negative voltages from positive ones with less effort (ICL7660).
Why not try one of those ?

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 26, 2018, 12:54:45 am**

Quote from: Wolfgang on August 25, 2018, 01:00:04 pm

> Hi,
>
> you probably know that there are specialized ICs to create negative voltages from positive ones with less effort (ICL7660).
> Why not try one of those ?

This is good advice. I suggested the CMOS 555 because they are widely available.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 26, 2018, 12:58:03 am**

Quote from: t1d on August 25, 2018, 06:11:09 am

> I have finished the rough draft, of the Dual Supply model. Here is the pfd.
>
> Snip ...
>
> I still need lots of help with proofing, on both models. Please, please, oh please and thank you. <grin>

There is a mistake in the VAC power supply option:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=506786;image)

You can't use the LM7909 to generate a negative voltage from a positive input.

The rectifier section should look like this:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=506792;image)

Two half-wave rectifiers, one is positive, the other negative.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 26, 2018, 01:13:23 am**

> Quote from: Jay_Diddy_B on August 26, 2018, 12:58:03 am
> There is a mistake in the VAC power supply option:

Well, I should be embarrassed... But, having no sleep, in 36 hours, I think I won't go to the bother.<grin>

I will get this cleaned up.

I have been using the time to work on my version of the project. I have the schematic and a rough draft, of the PCB. I will post it, just for fun, in a little bit...

Thanks!

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 26, 2018, 03:35:42 am**

> Quote from: t1d on August 26, 2018, 01:13:23 am
> I have the schematic and a rough draft, of the PCB. I will post it, just for fun, in a little bit...

Jay, thanks for proofing the schematic and catching the issue. Here is the corrected version, of the VAC Option. I will post the KiCad files, as soon as I hear back, on my other questions.

Posting my personal version will have to wait, I think...

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kleinstein** on **August 26, 2018, 07:20:54 am**

> Quote from: t1d on August 26, 2018, 03:35:42 am
> Quote from: t1d on August 26, 2018, 01:13:23 am
> I have the schematic and a rough draft, of the PCB. I will post it, just for fun, in a little bit...
> Jay, thanks for proofing the schematic and catching the issue. Here is the corrected version, of the VAC Option. I will post the KiCad files, as soon as I hear back, on my other questions.
>
> Posting my personal version will have to wait, I think...

There is a wrong connection from output to shunt in the lower right corner of the circuit drawing.

For current measurement, one should average the voltage over the different parts, the current sharing may not be very accurate (good enough for sharing the heat, but not for measurement. So it would be 2 or 4 resistors to the measurement output.

The choice of OPs mentioned here is odd:
The LT1013 is rather slow. It can be an option for a more static load with single supply.
The TL084 has quite some offset, drift and noise. The offset can cause quite some trouble at the low end, when due to the offset only 1 channel might be active. So at least in this case one should have the option to disable all but one channel when using only a small current.

When making a board also be aware of the odd pin-out of the SO8 version of the LT1013.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 26, 2018, 08:56:12 am**

> Quote from: Kleinstein on August 26, 2018, 07:20:54 am
> There is a wrong connection from output to shunt in the lower right corner of the circuit drawing.

Here is a picture:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=506960;image)

Remove the connection marked with the X.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 26, 2018, 09:01:02 am**

> Quote from: Kleinstein on August 26, 2018, 07:20:54 am
> For current measurement, one should average the voltage over the different parts, the current sharing may not be very accurate (good enough for sharing the heat, but not for measurement. So it would be 2 or 4 resistors to the measurement output.

The current monitor can be improved by averaging the voltage across the source resistors like this:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=506966;image)

or (the same thing)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=506972;image)

The resistor should be 50 \$\Omega\$ x number of output stages. With four output stages use 200 \$\Omega\$ resistors.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 26, 2018, 09:05:03 am**

Hi,

There is still an error in the VAC power supply option:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507008;image)

Make the changes shown in red.

Regards,
Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 26, 2018, 09:18:50 am**

Quote from: Kleinstein on August 26, 2018, 07:20:54 am

> The choice of OPs mentioned here is odd:
> The LT1013 is rather slow. It can be an option for a more static load with single supply.
> The TL084 has quite some offset, drift and noise. The offset can cause quite some trouble at the low end, when due to the offset only 1 channel might be active. So at least in this case one should have the option to disable all but one channel when using only a small current.
>
> When making a board also be aware of the odd pin-out of the SO8 version of the LT1013.

This is the most interesting part. The dual supply configuration allows a wide selection of op-amps to be used as it removes the need for rail-rail input or output.

I use the LT1013 for a lot of modelling because the LT1013 is very close to the LM324, except the LT1013 has better offset etc.

I would actually use the quad LT1014 if built the circuit. Standard quad op-amps will fit on the LT1014 footprint if pin 1 is lined up.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507014;image)

This model shows the same load circuit with the LT1013 (GBW 1MHz), the LT1801 (GBW 80 MHz) and voltage controlled voltage source (infinite GBW):

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507020;image)

If I compare the LT1013 versus the LT1801:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507026;image)

The loop bandwidth is 40kHz for the LT1013 and 60 kHz for LT1801. In both cases the phase margin is greater than 80 degrees.

or the LT1013 versus voltage controlled current source:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507032;image)

The voltage controlled voltage source could be considered an 'ideal op-amp'. Again there is very little difference between the LT1013 and the ideal op-amp.

The circuit is dominated by the feedback network around the op-amp and the MOSFET input capacitance and the gate resistor. See earlier in this thread for more analysis.

I see that the op choice has little impact, as it should on the small signal behaviour.

The LT1013 (LT1014) is good enough and the offset performance is desirable.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 26, 2018, 06:32:04 pm**

Quote from: Jay_Diddy_B on August 26, 2018, 09:18:50 am

> I would actually use the quad LT1014 if built the circuit. Standard quad op-amps will fit on the LT1014 footprint if pin 1 is lined up.

From when I was shopping for op amps, I looked at the LT1013/4 and TL074 data sheets. I remember seeing that, for some characteristic, that I now forget, there is some significant difference between the 13 and 14 specifications... Sorry to forget the exact item. But, if someone wants to use that brand, they should take a look... As I recall, the 13 had the better spec, for whatever it was...

I might take a look for it, later... But, I wanted to mention it, now, so, if someone was about to place an order, they could have this heads-up...

Thanks, soooo much, to everyone, for the notes, the schematic about... I am going to work on making those changes, now. I hope to post back, later in the day.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 26, 2018, 07:43:27 pm**

Here is the updated KiCad Schematic pfd.

Jay, I will leave it up to you (please,) to say when the schematic is complete. Once we do, you may want to add it to your very first post, for ease of finding it?

Jay, your making of handwritten notes, in red, on the schematic, is an excellent method of communication. Me grok. <grin> Please continue to include them...

Items from my prior posts, that we may (or may not?) want to consider, for finishing the schematic.

Quote from: t1d on August 25, 2018, 06:11:09 am

> Please clarify the suffix, of the values, of these resistors. I could guess, but that is dangerous. Such as R19 is a 51R resistor... These component reference numbers are for the Single Supply model, that I posted above, but they need to be updated in the Dual Model, too. *EDIT: These resistors have been renumbered, in the Dual Supply Model, because of changes. Be sure to refer to the Single Supply Model pdf.*
>
> R1 100
> R2 0.1
> R20 100
> R21 0.1
> R23 3300
> R4 3300
> R5 0.1
> R9 2.2
>
> Would it be okay, to bread board the circuit? Or, will there be issues... Voltage, amperage, capacitance? I would think that I would not want to run it above 1v/1a. If I can Bbd it, I will order the exact parts, from Mouser.
>
> At the risk of adding project creep, I have two questions:
> 1) If I wanted to add a switch, to bring in, or cut out, the oscillation section, without spinning the knob, where would it go?
> 2) If using an independent function generator would add functionality greater than the on-board oscillator, where, in the circuit, should it be injected? Just use the type jack that breaks the circuit? What issues might there be?
> 3) If two dual op amps are used, I would think that any unused units might be used to create a sign wave, too, in the oscillator section.
>
> Non-creep question:
> Would there be any great advantage to using ten-turn pots?
>
> I still need lots of help with proofing, on both models. Please, please, oh please and thank you. <grin>

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 26, 2018, 09:29:45 pm**

I believe that the difference I noted, between Lt1013/14 models was this...

	LT1013CDR	Other Models
Temp Max	70*C	125*C
Temp Min	0*C	-55*C
Input V Offset	0.3mV	0.8mV

Given that either operational temperature range is acceptable, I take it that the lesser 0.3mV input voltage offset, of the LT1013CDR, would be preferable to the typical 0.8mV, of the other models. Correct?

How important would the better input voltage offset advantage be, in a cost/benefit analysis? LT1013CDR/Dual @ $1.91/unit needs more PCB real estate. LT1014/Quad @ $5.02; YMMV by footprint. For me, at 5 MOSFETS, 4 x $1.91 < $2 x 5.02 and real estate doesn't matter, because 100mm x 100mm is a flat price.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 26, 2018, 11:49:55 pm**

Quote from: t1d on August 21, 2018, 04:57:24 am

> Snip ..
>
> Please clarify the suffix, of the values, of these resistors. I could guess, but that is dangerous. Such as R19 is a 51R resistor...
>
> R1 100
> R2 0.1
> R20 100
> R21 0.1
> R23 3300
> R4 3300
> R5 0.1
> R9 2.2

Resistor values given without m (milli), k (kilo) or M (Meg) are in Ohms, no suffix.

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 27, 2018, 12:00:07 am**

Quote from: t1d on August 26, 2018, 09:29:45 pm

> I believe that the difference I noted, between Lt1013/14 models was this...
>
	LT1013CDR	Other Models
Temp Max	70*C	125*C
Temp Min	0*C	-55*C
Input V Offset	0.3mV	0.8mV

> Given that either operational temperature range is acceptable, I take it that the lesser 0.3mV input voltage offset, of the LT1013CDR, would be preferable to the typical 0.8mV, of the other models. Correct?
>
> How important would the better input voltage offset advantage be, in a cost/benefit analysis? LT1013CDR/Dual @ $1.91/unit needs more PCB real estate. LT1014/Quad @ $5.02; YMMV by footprint. For me, at 5 MOSFETS, 4 x $1.91 < $2 x 5.02 and real estate doesn't matter, because 100mm x 100mm is a flat price.

There are some industry standard pinouts for dual and quad op-amps.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507536;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507542;image)

These pinouts are widely used and there are many parts that will fit a board designed with these pinouts.

The LT1013 uses a non-standard footprint:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507548;image)

This is not really a good choice, because it means you can't try other op-amps.

The LT1014 use a modified footprint:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507554;image)

There are two extra pins, no connect (nc) pins, but this causes the numbering to change. If you use this footprint, the LT1014 can be used and other parts with the standard footprint will also work.

The full scale voltage on the shunt resistor is 250mV so at 10% of full scale, 25mV, a 0.3mV offset is a little over 1% error. This give you an idea of the sensitivity to the offset voltage Vos.

A total supply voltage >24V is required. The GBW should be 1 MHz or greater.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 27, 2018, 01:31:50 am**

Quote from: Jay_Diddy_B on August 27, 2018, 12:00:07 am

> There are some industry standard pinouts for dual and quad op-amps.
> These pinouts are widely used and there are many parts that will fit a board designed with these pinouts.
> The LT1013 uses a non-standard footprint:
> This is not really a good choice, because it means you can't try other op-amps.
> The LT1014 use a modified footprint:
> There are two extra pins, no connect (nc) pins, but this causes the numbering to change. If you use this footprint, the LT1014 can be used and other parts with the standard footprint will also work.

E-x-c-e-l-l-e-n-t (cautionary)information..

Quote from: Jay_Diddy_B on August 27, 2018, 12:00:07 am

> The full scale voltage on the shunt resistor is 250mV so at 10% of full scale, 25mV, a 0.3mV offset is a little over 1% error. This give you an idea of the sensitivity to the offset voltage Vos.

For my personal need and purpose... to build a good, daily, bench load... and not to do further experimentation, I think I will go with the LT1013CDR.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 27, 2018, 01:34:26 am**

Quote from: Jay_Diddy_B on August 26, 2018, 11:49:55 pm

> Resistor values given without m (milli), k (kilo) or M (Meg) are in Ohms, no suffix.

I like an "R," but that's just me... So, is it okay to release the KiCad File?

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 27, 2018, 02:02:37 am**

I know that the shunt resistors need to be 1%, IIRC. Are there any other special components requirements? High wattage resistors, etc... Thanks

---

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 27, 2018, 02:48:58 am**

T1d,

Quote from: t1d on August 27, 2018, 01:34:26 am

> So, is it okay to release the KiCad File?

You can do whatever you want. My recommendation is that you wait until you have finished and that you know that it works properly.

I have released schematic in the past, that I haven't built but I have marked them as such.

Quote from: Jay_Diddy_B on December 02, 2013, 11:03:30 pm

> Hi,
> I am going to share the design and construction of an ESR Meter adapter design and construction. The plan is to end up with a design that is similar in appearance to Dave's ucurrent that will allow a DMM to be used as an ESR meter. ~~**At this point in time I have not built the circuit.**~~
>
> The requirements are:
>
> Snip ...
>
> Jay_Diddy_B

I crossed out the warning after the circuit was built and tested.

You may want to swap the op-amps around during layout, to get a nice design.

It would be courteous to reference this thread on a schematics and board artworks that you release.

example:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507611;image)

My original artwork is shown in this post:

https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/msg630287/#msg630287 (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/msg630287/#msg630287)

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 27, 2018, 02:53:20 am**

> Quote from: t1d on August 27, 2018, 02:02:37 am
> I know that the shunt resistors need to be 1%, IIRC. Are there any other special components requirements? High wattage resistors, etc... Thanks

All the resistors can be 1%. The cost of 1% versus 5% is minimal.

Most parts can be 0603, 0805 or 1206 whatever you are comfortable working with. You need 2512 resistors for the shunts.
The 2.2uF should be 100V 1210 size.

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 27, 2018, 03:09:40 am**

Thank you!

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 27, 2018, 03:33:53 am**

> Quote from: Jay_Diddy_B on August 27, 2018, 02:48:58 am
> I have released schematic in the past, that I haven't built but I have marked them as such.

This is what I have/am intended to do. I just would have more confidence, in my work, if there was a consensus that it is correct. Not as to design operations, but as to circuit completeness and correctness. My desire is to contribute to the DIY community, by helping with creating a KiCad file, which should have wide utility. I don't feel any obligation, to warrant the design... nor should you, given open source nature, of this thread.

> Quote from: Jay_Diddy_B on August 27, 2018, 02:48:58 am
> It would be courteous to reference this thread on a schematics and board artworks that you release.

Yes, I have remembered to do that. The schematic already has your name/handle credit. I will add the link info. My board layout already has both, but I really hadn't intended to release it, because it is tweaked, to my purposes. But, if someone wants it, I would share it, readily...

Title: **Re: Dynamic Electronic Load Project**
Post by: **JS** on **August 27, 2018, 04:02:45 am**

> Quote from: t1d on August 27, 2018, 03:33:53 am
> > Quote from: Jay_Diddy_B on August 27, 2018, 02:48:58 am
> > I have released schematic in the past, that I haven't built but I have marked them as such.
> >
> > This is what I have/am intended to do. I just would have more confidence, in my work, if there was a consensus that it is correct. Not as to design operations, but as to circuit completeness and correctness. My desire is to contribute to the DIY community, by helping with creating a KiCad file, which should have wide utility. I don't feel any obligation, to warrant the design... nor should you, given open source nature, of this thread.
> > Quote from: Jay_Diddy_B on August 27, 2018, 02:48:58 am
> > It would be courteous to reference this thread on a schematics and board artworks that you release.
> >
> > Yes, I have remembered to do that. The schematic already has your name/handle credit. I will add the link info. My board layout already has both, but I really hadn't intended to release it, because it is tweaked, to my purposes. But, if someone wants it, I would share it, readily...

If I were to build one of this which I'd probably like to, I'd include, as I did in mine, a thermal shout down. It's just matter of placing a LM35 on the heatsink and a comparator with some hysteresis shouting the thing down for a certain reference voltage. That makes it much more robust if something goes wrong I you don't catch in on time, as you are centered on other parts of the test other than the dummy load surviving it, like the DUT.

JS

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 27, 2018, 10:55:56 am**

Latest Dual Model schematic... I think I just added some notes...

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=507725)

Title: **Re: Dynamic Electronic Load Project**
Post by: **Mihkel** on **August 27, 2018, 01:19:23 pm**

There is one important connection missing, between D1 cathode and V1 input.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 28, 2018, 12:29:38 am**

> Quote from: Mihkel on August 27, 2018, 01:19:23 pm
> There is one important connection missing, between D1 cathode and V1 input.

Excellent catch, Mihkel.

It was easy, to process, quickly. I think I will post the KiCad files, tomorrow night. It would be great, if everybody would give the schematics one more look. Please and thank you...

You will recall, that I also made the schematic for the Single Supply model. It was only made, to the specs, of its model. That means that it does not have all the bells and whistles, that we put into the Dual Model. But, the user is welcome to finish it, as they like.

Jay, speaking of the Single Supply model, it was spec'd with just one 51 Ohm resistor, going to the Meter Output. Should it be one 100 ohm resistor, for each of the two MOSFETs?

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508178)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508181)

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 28, 2018, 02:11:41 am**

Hi,

Here are some more corrections.

On the single supply version, you have the current monitor on the Drain of the MOSFET it should be the Source:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508217;image)

On the double supply version, the scaling factor is wrong. You have 8 x 0.2 \$\Omega\$ resistors in parallel so the scaling is 25mV / A:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508223;image)

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 28, 2018, 02:33:54 am**

Hi,

Here are some comments on the input connector:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508247;image)

The schematic shows the part as conn_0.1x2, which suggest this type of connector:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508253;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508259;image)

The current rating of the connector is determined by the size of the wire you can attach to the mating part. This is limited to 22 awg (0.32 mm^2) which is about 1A.

I would use something else.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 28, 2018, 02:46:48 am**

T1d,

You need to sort out the op-amp pin numbers.

In an earlier message you had decided to use the LT1013 op-amp with the non-standard pin connections:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508283;image)

But you are showing the, standard pinout, quad op-amp on the schematic:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508289;image)

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 28, 2018, 03:20:17 am**

Good points, Jay.

Quote from: Jay_Diddy_B on August 28, 2018, 02:11:41 am

> On the double supply version, the scaling factor is wrong. You have 8 x 0.2 \$\Omega\$ resistors in parallel so the scaling is 25mV / A:

Actually, there are 5 pairs of resistors, at 0.2. I included five MOSFETS. I just replicated the MOSFET nodes, from the simulation circuit. I don't know how to do the math, to get 50 \$\Omega\$, but, if you want me to use it as a learning exercise and correct it, I would be glad to do that. We probably need to add a note, to the schematic, about adjusting the resistor values, based on the number of MOSFETs.

Quote from: Jay_Diddy_B on August 28, 2018, 02:33:54 am

> Here are some comments on the input connector: The schematic shows the part as conn_0.1x2, which suggest this type of connector: The current rating of the connector is determined by the size of the wire you can attach to the mating part. This is limited to 22 awg (0.32 mm^2) which is about 1A. I would use something else.

One of the great advantages of KiCad, is that the schematic symbol does not dictate the board component. This is a good example. The schematic specifies what I thought to be a generic connector, of any type, having two pins. For the unit I will be building, I will be using banana jacks. I will assign the banana jack footprint to the generic two-pin connector symbol, with the component assignment function. But, not everyone may want banana jacks.

Because you have made me aware that this particular two-pin connector symbol actually specifies the connector you showed, I will see if there is not a banana-style connector symbol. Or, would you think some other type connector would be better?

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 28, 2018, 04:19:22 am**

I searched for a banana jack symbol, until I realized how long I had been at it... There's a hour, I'm not getting back. This is the closest symbol I could find. It's easy enough to create the symbol, in KiCad, if this symbol (doughnut) will do. Thoughts?

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508382)

EDIT: Here's my KiCad version. Even if we don't use it, for this project, I need it, for my general library of symbols.
(https://www.eevblog.com/forum/index.php?action=dlattach;topic=20497.0;attach=508409;image)

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 28, 2018, 06:47:03 am**

This is the correction, to the Single Supply model. The reference numbers will be R5 and R6.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508463)

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 28, 2018, 08:41:38 am**

Quote from: Jay_Diddy_B on August 28, 2018, 02:46:48 am

> T1d,
>
> You need to sort out the op-amp pin numbers.
>
> In an earlier message you had decided to use the LT1013 op-amp with the non-standard pin connections:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508283;image)
>
> But you are showing the, standard pinout, quad op-amp on the schematic:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508289;image)
>
> Regards,
>
> Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 28, 2018, 09:01:09 am**

Quote from: Jay_Diddy_B on August 28, 2018, 02:46:48 am

> T1d,
>
> You need to sort out the op-amp pin numbers.
>
> In an earlier message you had decided to use the LT1013 op-amp with the non-standard pin connections:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508283;image)
>
> But you are showing the, standard pinout, quad op-amp on the schematic:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508289;image)
>
> Regards,
>
> Jay_Diddy_B

Doh! You are exactly correct. In the beginning, I was trying to decide whether to use quads, or duals. Looking at the device specifications, between the LT1013 and the LT1014, they are the same, IIRC... (except for the LT1013CDR, as previously discussed...) So, my thinking was that quads save real estate and I put in the quad symbol. But, as I went back and labeled the components from your model, I dropped the ball and assigned LT1013, as the device reference.

So, we have a choice, here... I can just change the labeling to "LT1014." That's the easiest solution. Or, if you prefer the dual, I don't mind making the pin changes.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kean** on **August 28, 2018, 11:49:14 am**

Another minor error - the 100R resistor for the monitor output (near R21 labelled R?) connects to ground instead of the junction of R21 and M2 source pin.

Quote from: t1d on August 28, 2018, 06:47:03 am

> This is the correction, to the Single Supply model. The reference numbers will be R5 and R6.
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508463)

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 28, 2018, 02:31:25 pm**

Quote from: Kean on August 28, 2018, 11:49:14 am

> Another minor error - the 100R resistor for the monitor output (near R21 labelled R?) connects to ground instead of the junction of R21 and M2 source pin.

Nice one, Kean! The two unnumbered resistors have been assigned as R5 and R6. Here's your fix:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508778)

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 28, 2018, 02:45:29 pm**

I went ahead and built the SOIC-8 pinout, for the LT1013 symbol. Making the needed Dual-or-Quad Op Amp change will be easy. Just let me know, which way to go. If we go with the Quad LT1014, I will try to remember to include the LT1013 symbol, in the files I post, so individuals can easily change the design.

The Quad LT1014 has a common footprint to the TL074. So, if an individual wants to use the LT1014, they just place the TL074 and change its name.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508829)

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kleinstein** on **August 28, 2018, 03:00:46 pm**

For the FET drivers, there is likely not much space to save from using a quad version (e.g. LT1014), because the power MOSFETs need so distance in between anyway. So it is 2 extra pins versus longer tracks. The square wave generator part does not need a precision OP - here an LM358 / 741 could be good enough. It's mainly the output drivers that need to be precision due to the low voltage at the shunts.

For the MOSFET choice, I would be careful with IRFZ20: The DS (from vishay) shows an DC FBSOA, but it is a reasonable modern switching MOSFETs for just 50 V. The FBSOA might be just calculated from transient thermal response and not include possible thermal instability.

The more sensible alternative would be to consider an TO247 case type like IRFP250. These could handle more power in a single chip and thus less channels needed. The higher voltage rating and thus low trans-conductance also helps with thermal stability.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 28, 2018, 03:23:45 pm**

Quote from: Kleinstein on August 28, 2018, 03:00:46 pm

> The square wave generator part does not need a precision OP - here an LM358 / 741 could be good enough. It's mainly the output drivers that need to be precision due to the low voltage at the shunts.

There tons of function generators circuits built on the XR2206 and the AD8038, too.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kean** on **August 28, 2018, 04:04:01 pm**

Nope, that fix is not going to work.
R21 is a shunt resistor and has to go between the FET source pin (M2 pin 3) and GND.
Your monitor output then needs to average out the voltages across all N shunt resistors by having an Nx50 ohm resistor from the high side (non grounded side) of each shunt resistor.
You've got the configuration correct with M1, R2, and R6 - but it is totally messed up around M2.

Quote from: t1d on August 28, 2018, 02:31:25 pm

> Quote from: Kean on August 28, 2018, 11:49:14 am
>
> > Another minor error - the 100R resistor for the monitor output (near R21 labelled R?) connects to ground instead of the junction of R21 and M2 source pin.
>
> Nice one, Kean! The two unnumbered resistors have been assigned as R5 and R6. Here's your fix:
>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508778)

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 28, 2018, 05:09:37 pm**

Thank you, for seeing this through, Kean. Hopefully, this better. If not, please keep after me. I appreciate your help.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=508871)

Title: **Re: Dynamic Electronic Load Project**
Post by: **hugo** on **August 28, 2018, 11:52:21 pm**

Hi Jay,

How about replacing the IRF530/IRFZ20 mosfets with some low side power switch like BTS117/BTS133 to make it bulletproof, what do you think, will those work?

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 29, 2018, 01:13:41 am**

Quote from: hugo on August 28, 2018, 11:52:21 pm

> Hi Jay,
>
> How about replacing the IRF530/IRFZ20 mosfets with some low side power switch like BTS117/BTS133 to make it bulletproof, what do you think, will those work?

The issue with the BTS117/BTS133 is that they are SMD packages with no tab to bolt to a heatsink.

If I was to buy MOSFETs for a load I would look at PSMN4R8-100PSE from Nexperia (NXP) These have extended SOA.

Link: https://www.nexperia.com/products/mosfets/power-mosfets/PSMN4R8-100PSE.html (https://www.nexperia.com/products/mosfets/power-mosfets/PSMN4R8-100PSE.html)

I also like the power dissipation protection implemented in this load:

https://www.eevblog.com/forum/projects/complete-project-constant-current-dummy-load/ (https://www.eevblog.com/forum/projects/complete-

project-constant-current-dummy-load/)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=509141;image)

This circuit would work better if the transistors were replaced by a transistor array like the LM3046. If you use the transistor array the substrate must be connected to the most negative part of the circuit.

I would need to do some analysis to see if the discrete transistors are good enough.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 29, 2018, 05:28:46 am**

Quote from: Jay_Diddy_B on August 28, 2018, 02:46:48 am

> T1d,
> You need to sort out the op-amp pin numbers.
> In an earlier message you had decided to use the LT1013 op-amp with the non-standard pin connections:
> But you are showing the, standard pinout, quad op-amp on the schematic:
> Regards,
> Jay_Diddy_B

Jay, I noticed that the original schematic release used Quad TL074's. This gives me understanding, of the error I made, in crossing up the Dual LT1013 identifier, with the Quad LT1014 footprint.

Now that I understand what occurred, the solution becomes clear. I will label the existing Quad schematic, with "LT1014," because the pinout is the same as the TL074. And, I will provide an additional schematic, using the LT1013 and its non-standard pinout.

Here is an example of both. These have my dual ac power supply setup and the LT1019 V Ref, that I have in stock. My final release will include both, of these options. Plus, I will include the battery power supply and the LT6656 V Ref options.

I have inserted 1x3 connectors, between the MOSFETs and their Op Amps. I know that, because of the power involved, I should not use connectors. I will just use the holes to hard wire my control board, to my MOSFET board. I am still looking for the wire pad symbol, to label these holes, correctly.

If you would provide me with the formula, to calculate the resistor value, needed for changes in the number of MOSFETs, I would like to include it, as a note, on the schematic. Then, people can scale the number of MOSFETs, as they like.

Thanks.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=509237)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=509240)

Title: **Re: Dynamic Electronic Load Project**
Post by: **hugo** on **August 30, 2018, 01:34:45 am**

Quote

> The issue with the BTS117/BTS133 is that they are SMD packages with no tab to bolt to a heatsink.

They can be bolted to a heatsink, I guess:

https://www.digikey.ca/product-detail/en/infineon-technologies/BTS117BKSA1/BTS117BKSA1-ND/5410153 (https://www.digikey.ca/product-detail/en/infineon-technologies/BTS117BKSA1/BTS117BKSA1-ND/5410153)

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **August 31, 2018, 01:23:42 am**

Jay,

Is the information contained in the attachment what I need to learn, to be able to scale the value of the MOSFET heat sinks, for a given number of MOSFETs? I just don't want to be learning the wrong thing.

The KiCad files are just about ready to go. I just have a few more tweaks and I want to add the heat sink formula, as a note.

Thanks

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 01, 2018, 02:22:57 am**

This is the current revision, of the Single Model. It had the same symbol name/symbol icon issue. I cleaned that up, by assigning the LT1014 name, to match the existing quad symbol.

I have finished all the KiCad schematics that I intended to contribute. The only other thing that I would like to do is to add a note about how to calculate the sink resistor values, for changes in the number of MOSFETs.

What about spreading the current through multiples of these resistors. Do we need to add a comment about that? I see that some of the designs use one and others use two.

In reading back through the entire thread, I realized that a good many of the questions, that I was asking repeatedly, and that had gone without response, were answered, in prior posts. My sincere apologies, for that. However, I did not see the answer, regarding the sink resistor calculations. If I missed that, in the posts, please direct me to the post number.

As this version, of the Single Model, has undergone significant change, I would appreciate help in proofing it. Thank you, in advance, for your help. In the meantime, I will be finishing the release notes.

Hmm... I can't get the pdf image, to open, in the text frame. I am using the image function and inserting the file location link, but no joy? (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=512048)

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kleinstein** on **September 01, 2018, 08:23:36 am**

The shown circuit is odd in that the upper 2 MOSFETs get a 10 times higher current than the lower ones. The divider at the OPs output is only for the lower ones. There is also no need to use the same type of OP for the whole circuit.

With the divider before the output stages it is possible to use a much lower grade OP (e.g. LM358) for the initial part. It does not make sense to include the TL084 option in the single supply version - it just won't work. Something like an OPA4171 would be a suitable single supply replacement.

The IRFZ20 likely not a viable option for linear operation :horse:

For the parallel connection of the shunt resistors, it can be an option, but it does limit the accuracy. Also keep in mind that for accurate current shunts one has to operate them at well below nominal power: at nominal power the self heating is usually too high and causes excessive change in the resistor value. The 0.1 Ohms shunts should operate at something like up to 1 A and thus 100 mW. Still it would make sense to choose resistors that are good for a 0.5 or 1 W power rating. 250 mW types are too small - even with just 100 mW used.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 02, 2018, 03:53:46 am**

Quote from: Kleinstein on September 01, 2018, 08:23:36 am

> For the parallel connection of the shunt resistors, it can be an option, but it does limit the accuracy. Also keep in mind that for accurate current shunts one has to operate them at well below nominal power: at nominal power the self heating is usually too high and causes excessive change in the resistor value. The 0.1 Ohms shunts should operate at something like up to 1 A and thus 100 mW. Still it would make sense to choose resistors that are good for a 0.5 or 1 W power rating. 250 mW types are too small - even with just 100 mW used.

Good point. I will add this, as a note, to the schematics.

As for the other matters, they are above my pay grade. But as for the upper MOSFETs seeing more current, aren't all the MOSFET nodes connected in parallel?

Do you know how/formula, for calculating the MOSFET's sink resistor values, based on the number of MOSFETs used? That information is all that I need, to finish the schematics, for publication. I want to publish them, this weekend, and move on to other matters.

Thanks!

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kleinstein** on **September 02, 2018, 08:29:31 am**

In the last schematics, there is a mistake (obvious drawing fault) to give a higher control voltage to the upper stages.

How much voltage to use for the resistor for current regulation is not easy to answer. Using a small resistor and thus only low voltage (e.g. 100 mV) has the advantage of a lower minimum working voltage and also keeps the power dissipation low. A larger resistor can give lower noise and drift, as the voltage is larger and thus easier to handle. This is especially important if only a small fraction of the maximum current is used. So the choice is a kind of compromise, likely with something like 200 mV ...500 mV at full current. So the shunts need to be high power: depending on the value one could consider 5 W or even higher power rating. So SMD may not be the best choice. There is a reason the commercial electronic loads often use quite bulky wire type resistors.

One could slightly ease on the problem by a kind of range switching: for low current use only 1 power stage, possibly even a special stage made for lower currents.

The TL084 is cheap, but also rather poor, when it comes to DC and low frequency noise performance. The OPA4171 (alternative slightly lower grade TLV4171) is not that much more expensive. As the power stages are quite large anyway because of the heat sink, I would personally prefer dual or even single OPs anyway. Another affordable alternative would be RC4558 (dual) - still better noise and DC performance than a TL082. A better OP could permit saving on the shunt (smaller and lower power). So for a higher performance solution I would even consider something like OP27 or OPA211 - noise can matter.

For the power MOSSFETs, I would no take the risk with low voltage switching types. There is a real risk they could fail (usually short) due to internal thermal runaway. Something like an IRFP250 (TO247) is not that much more expensive and could replace about 2 of the TO220 ones.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 02, 2018, 11:29:38 am**

Quote from: Kleinstein on September 02, 2018, 08:29:31 am

> In the last schematics, there is a mistake (obvious drawing fault) to give a higher control voltage to the upper stages.

I thought the MOSFETs were all in parallel, at this point. Could you draw a picture, for me, like Jay did? Remember, I am still something of a noob, when it comes to design. That's why I have stayed out of the design conversations and made my contribution to only be the KiCad drawings.

Quote from: Kleinstein on September 02, 2018, 08:29:31 am

> How much voltage to use for the resistor for current regulation is not easy to answer.

I worded my question poorly. What I want to know is how to calculate the resistor value, for the MOSFET sinks, based on the number of MOSFETs and a 50mV (design spec) output.

The Spice model used two 0.2 ohm resistors, for two MOSFETS. I carried the two resistor design (and resistor value), over to the schematics, in which I had included 5 MOSFETs, to reach a 90 watt capacity (to handle testing of PSU designs up to 30v/3a.) Jay said that 0.2 ohms was incorrect, but he did not tell me the formula, to calculate the resitor values correctly. I want to add a note, to the schematics, as to how to calculate the resistor value, so that folks can change the number of MOSFETs, as they would like.

Quote from: Kleinstein on September 02, 2018, 08:29:31 am

> The TL084 is cheap, but also rather poor, when it comes to DC and low frequency noise performance.

As said, I will leave the design elements up to you guys. But, I hope to learn enough to join in, in the future.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kleinstein** on **September 02, 2018, 01:30:50 pm**

The choice if resistor value depends on the maximum set voltage and the maximum current per MOSFET. So the choice is the described design question: a low resistor allows cheaper low power resistors, but causes more noise and drift. So it depends on the priorities - there is no universal correct value.

The IRF530 and maybe IRFZ20 may be good for 1 A up to 30 V, maybe a little more. IF higher voltages can be present, a lower limit might be a good idea.

Assuming a 2,5 V reference (there are different voltage versions of the LT1019), the summing amplifier gain of 2 and with the divider 18K/2K the set voltage goes up to 0.5 V. So for a 1 A limit the shunts should be 0.5 Ohms or 2 of 1 Ohms in parallel. With 2 x 0.2 Ohms the current could go up to 5 A per FET and thus too high with more than about 10 V.

Besides the more obvious drawing error, there is a more general problem with the circuit: It lacks a circuit to limit regulator windup. So when sink circuit is powered before the source is connected, the OPs in the output stages will force the gate all the way up, because no current can flow. If than an external voltage source is connected, it takes some time for the MOSFETs to turn down - so quite a significant current puls could flow. depending on the power of the source this might even cause some damage.
It would be a good practice to turn off/down the current sink if the external supplied voltage drops too far (e.g. below 0.7 V).

Here the more obvious error in the last drawing (Yesterday at 04:22:57):

Title: **Re: Dynamic Electronic Load Project**
Post by: **Wolfgang** on **September 02, 2018, 01:51:23 pm**

... not powering up before a certain drain voltage is reached could result in a "stuck" situation when the power supply attached does not provide enough current to overcome this.
Another solution would be to limit gate voltage so that the MOSFETs are not driven into full saturation.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 02, 2018, 03:43:19 pm**

Thank you, Kleinstein, for the great explanation and the wonderful drawing. I have to admit, I am going to have to chew on the variables, a bit... lol... In the meantime, I will straighten up the schematics.

Super information, Wolfgang!

Thank you, both.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 02, 2018, 04:15:34 pm**

Here's the fix... Thanks!

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=513323)

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kleinstein** on **September 02, 2018, 04:30:00 pm**

Quote from: Wolfgang on September 02, 2018, 01:51:23 pm

> ... not powering up before a certain drain voltage is reached could result in a "stuck" situation when the power supply attached does not provide enough current to overcome this.
> Another solution would be to limit gate voltage so that the MOSFETs are not driven into full saturation.

It would be a rather unusual supply that has trouble to overcome the minimum start voltage with the load turned off. The danger is more in getting an oscillation where the voltage breaks down below the limit when the load is turned on.

Limiting the gate voltage would help a little: it makes the reaction faster and could act as an additional current limit if set accurately.

A good solution would be to limit the effective resistance to something like twice the shunt value. This would be by limiting the set voltage to something like twice the output voltage minus a small offset. So at very low voltages the load would switch over from a constant current mode to a kind of constant resistance mode, not allowing to go below about 2 times the shunts.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 03, 2018, 07:41:05 am**

Hi, contributors,

In my mind, the development of the KiCad files had reached the point of only needing a comment, as to how to calculate the meter output resistors. I was way, way, over thinking that. The information had, previously, been provided. My apologies, for repeatedly asking for the formula. The light finally came on!

So, as promised, here is the (first) release, of the KiCad files. I need some help, please...
- Open the KiCad files, download my custom symbol library, and add the custom library, to your library preferences. Make sure that the schematics include the custom symbols, when opened. See the following post, for files.
- I have made additional (minor) changes, to all three schematics. Please proof them... hard. Pdf's provided, below.
Thank you, for this help.

The meter output is specified as 50mV/A. I would like to have a V/A ratio of 10:1. That way, whatever my multimeter volts reading, that is the number of amps being sunk. Is there any problem with using a 9K/1K resistor divider network?

A note to non-KiCad users, making use of the pdf files:
You are free to make any and all use, of the files. However, in doing so, you release me from absolutely all liability. I am not an electrical engineer. I do not guarantee that there are no mistakes. You must verify absolutely everything, for yourself. I did not design the operational circuit.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 03, 2018, 07:46:57 am**

In appreciation, of Jay_Diddy_B's ingenious design, hard work and gracious sharing, of his Dynamic Electronic Load Project, I am contributing the KiCad files, for his schematics. KiCad is a powerful, FREE program, to make schematics and board designs. Find it here: **[Domain Removed]** It will be very easy, to create PCB boards, with the files.

What Is included
Circuit development branched into two divisions; the original Dual (Power) Supply model and the Single (Power) Supply model. The Dual Model schematic was further developed to use a common pinout quad op amp (the LT1014, or the TL074) and the dual LT1013 op amp (having a unique pinout.) Therefore, there is a total of three schematics.

The schematics are ready for you to use, as is. You will only need to:
- select your choice of power models,
- select your choice of op amps,
- select your choice of power supply and remove the one you will not be using,
- select your choice of voltage reference and remove the one you will not be using,
- select your components and
- create your own PCB board design.

Feel free to customize. Use components that you like, or have in stock. The selection of the component types, for the board, is done, with the CvPCB function, found on the schematic page. Depending on your selection, from the Power Supply and Voltage Reference options, you may have to relocate the power flags. Whatever you plan to do, you will be miles ahead, using these files.

A note on customization… Changes in components will effect performance; you need to know what you are doing, or get help. If you change the number of MOSFET nodes, you will need to calculate the new value, for their meter output resistors. If you want to use a LT1013 dual op amp, on the Single Model, you will have to change the connections, to the new op amp's pinout, as the does not have a common pinout.

The wattage, seen by the shunt resistors, can be spread over parallel resistors, but it does limit the accuracy. For accurate current shunts, one has to operate them at well below nominal power; at nominal power the self heating is usually too high and causes excessive change in the resistor value. The 0.1 Ohms shunts should operate at something like up to 1 A and thus 100 mW. Still it would make sense to choose resistors that are good for a 0.5 or 1 W power rating. 250 mW types are too small – even with just 100 mW used. So, the value of parallel shunt resistors equals 0.1 x number of parallel shunt resistor used.

The value of the meter output resistor should be 50 ohms x number of output stages. For example, with four output stages use 200 ohm resistors.

A Note To SMD Users
All the resistors can be 1%. The cost of 1% versus 5% is minimal. Most parts can be 0603, 0805 or 1206 whatever you are comfortable working with. You need 2512 resistors for the shunts. The 2.2uF/100V should be 100V 1210 size.

A Note To Non-KiCad Users
KiCad allows the selection of component footprints, regardless of their schematic symbol. 01x03 connectors, labeled as "Wire Pads," are used here, to represent the selection of Wire Pads, on the PCB board. This method enables the use of separate Control and MOSFET PCB boards. However, these connections should be hard wired. Because of high power draws, connectors should not be used, to connect boards, to MOSFETs. If a single, Control-MOSFET board is used, remove these pads, from the circuit.

You are free to make any and all use, of the files. However, in doing so, you release me from absolutely all liability. I am not an electrical engineer. I do not guarantee that there are no mistakes. You must verify absolutely everything, for yourself. I did not design the operational circuit.

Edited by moderator 02/12/2021 - Removed reference to an old URL which is now potentially malicious.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kleinstein** on **September 03, 2018, 06:34:12 pm**

With the shunts at 0.1 Ohms, the meter output is 100 mV/A /N. Where N is the number of power stages. So the scaling of the output depends on the number of power stages. If the output is large enough, it would be acceptable to have a divider to adjust the range to an easier scale.

One would have to adjust the R18/R33 divider to something like 1:50 to limit the current to an acceptable level for the power stage.
With such a small shunt resistance it would be a good idea to have really low noise / low drift OPs. For the single supply version this could be something like OP213 or similar. The LT1014 might still be ok. However the TL074 is not working single supply :horse: and too noisy to be a practical choice with such a low shunt value.

The alternative would be keeping the 500 mV full scale range and this use something like 0.5 Ohms shunt, that than would need to be high power (like 5-20 W) types and thus likely THT form factor like a TO220, larger case cemented ones or bare wire shunts or quite some size.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 04, 2018, 05:03:49 am**

Excellent answer, as always, Kleinstein. I will continue to think on it.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 12, 2018, 12:30:35 am**

I have asked this before, but is there any reason that I could not use AD8032ANZ op amps and MTP3055VL MOSFETs, in Jays design? I have these, in stock. Thanks.

Title: **Re: Dynamic Electronic Load Project**
Post by: **bicycleguy** on **September 12, 2018, 01:58:43 am**

Quote from: BravoV on November 01, 2016, 03:03:12 pm

> ..snip snip..
> Jay, I have this 500V 46A "linear" mosfet IXTN46N50L (photo (https://www.eevblog.com/forum/chat/what-did-you-buy-today-post-your-latest-purchase!/?action=dlattach;attach=221228;image)) , datasheet (HERE (http://ixdev.ixys.com/DataSheet/89945a33-9e28-4242-b812-008abd738156.pdf)).
>
> It has a whopping gate capacitance (Ciss) at 7000 pF, just wonder if the circuit needs a major revision just to make it work as pulsing dummy load ? Say at "reasonable" rise & fall time for working at "common" power supply types. Also when running at the static continuous current, does need modification too ?

A while back in this thread Jay posted a lot of LTSpice work with the IXTN46N50L. Has anyone done anything with this? Seems like the package isolation and low RthJC and RthCS would make it easier to deal with than a bunch of parallel mosfets.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 12, 2018, 02:11:50 am**

As I recall, the author already had these, in his stocks. At $42/ea, at Mouser, it is out of my price range... I imagine that would be so, for most hobbyists. But, yes, you can sink more load, with fewer MOSFETs.

Title: **Re: Dynamic Electronic Load Project**
Post by: **bicycleguy** on **September 12, 2018, 02:17:48 am**

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t1d,
Seems like 1 IXTN could replace maybe 3 to 4 typ mosfets and associated parts, depending on what V and A you were looking for. I think it would be cheaper and easier.

edit: Also provides a way to mount your PWB ^-^

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 12, 2018, 02:26:36 am**

Easier, yes... But, the typical MOSFET needed is cheap... About $1/ea... If I had the money, I would surely do as you suggest. I have even been trying to figure out how to do it, on a retirement income budget. I haven't figured that out, yet.<g> There are some less expensive models of this type... $25/ea. So, do some looking...

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kleinstein** on **September 12, 2018, 04:00:21 pm**

The AD8032ANZ op amps and MTP3055VL are both poor choices:
the OP is very fast (which only makes life more difficult and is not needed) but has a rather large offset drift and like high LF noise. What is needed is more like a precision single supply OP with something like 1 MHz GBW, likely BJT based. So the LT1013 is a little on the slow side and still rather noisy and maybe on the low speed edge. If there is a negative supply available it would be something like an OP27 or ADA4075 that would work well. A good single supply choice could be the OP113 - though a little expensive. Depending on the frequency range one is interested in, one might also consider an AZ OP like AD8551 or MCP6V27. A low cost candidate might be TLC272 or TLV171, maybe even LM358.

The MTP3055 might have trouble with reliable operation due to SOA limitations :horse:.
A better choice would be something like IRFP250. It is more like looking for 200 V - 800 V MOSFETs, even if only 30 V are planed to use. A lower number of larger case FETs also reduces the number of OPs needed so that better quality ones could be used.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 12, 2018, 06:15:14 pm**

Great information, Kleinstein. I will investigate these parts.

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 14, 2018, 02:49:30 am**

Hi, Kleinstein,

I have been giving your suggestions consideration. They have good merit. I think it would be a good idea, for you to start a separate thread discussing
- the op amp selection factors
- the MOSFET selection factors
- and, what changes are needed (broadly,) in the circuit, based on these selections.
I know that I would be interested and, certainly, many others.

I looked at the IRFP250. It appears to be a good, reasonably priced candidate. What do you think of the Microchip MTP6002 op amp, as a match, for it? It might be best to answer, in the new thread, to keep from highjacking Jay...

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 17, 2018, 03:58:12 am**

A circuit to add a Function Generator Auxiliary Input to Jay's design can be found here, at Post #42...

https://www.eevblog.com/forum/projects/hacking-a-dynamic-electronic-load-circuit-to-use-an-external-function-generator/new/#new (https://www.eevblog.com/forum/projects/hacking-a-dynamic-electronic-load-circuit-to-use-an-external-function-generator/new/#new)

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **September 22, 2018, 04:22:12 am**

Here is another option, that I made, using TLV171 op amps and IRFP250 MOSFETS. The complete project release can be found here:
https://www.eevblog.com/forum/projects/electronic-load-project-ltv171-irfp250-with-kicad-files/msg1840739/#msg1840739 (https://www.eevblog.com/forum/projects/electronic-load-project-ltv171-irfp250-with-kicad-files/msg1840739/#msg1840739)

Title: **Re: Dynamic Electronic Load Project**
Post by: **jrsikken** on **September 24, 2018, 11:50:47 am**

Hi there,
I have designed an Arduino compatible electronic load and sold it to a hunderd people in the past year. You may want to look at my latest deisgn.

https://github.com/jrsikken/ElectronicLoadR3 (https://github.com/jrsikken/ElectronicLoadR3)

It uses the super robust BTS133 in TO-220 package. It has over voltage, over current, over power, ESD protection and thermal protection. It will save a lot of mosfet replacements. And a thermal safety control circuit.

In my circuit I had oscillation and so I had tuned the constant current circuit (opamp, mosfet, feedback circuit) to prevent oscillation but have fastest response. Now the current settles in 20us. Now it's so fast it supports pulses loads. For example the 550us wide 2A pulses from GSM can be simulated.

My design uses a DAC, the MCP4725 to generate an analog input for the constant current circuit. My embedded code can change DAC output voltage at max 5kHz. On this forum post http://www.stm32duino.com/viewtopic.php?t=1048 (http://www.stm32duino.com/viewtopic.php?t=1048) there is a guy who generated a 3kHz sine wave, with 60kHz sample rate. Basically it means you can program any waveshape from he MCU.

Title: **Re: Dynamic Electronic Load Project**
Post by: **GigaJoe** on **February 07, 2019, 05:15:20 pm**

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jrsikken,
I'm not certain C11 cap to load opamp are good idea, in addition to mosfet capacitance ... and C9 ... , seems all to fight for oscillation suppression

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Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **February 07, 2019, 11:35:09 pm**

Quote from: GigaJoe on February 07, 2019, 05:15:20 pm

> jrsikken,
> I'm not certain C11 cap to load opamp are good idea, in addition to mosfet capacitance ... and C9 ... , seems all to fight for oscillation suppression

I have had a look at the load designed by jrsikken. This is the circuit that GigaJoe is talking about:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=646757;image)

This is fairly conventional feedback around the op-amp IC1A. The note is interesting about the tuning.

The BTS133 Smart Sipmos device has a lot of extra 'stuff' in it:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=646763;image)

It has over current and over temperature protection which is good, but it also has circuits to limit dv/dt which is bad in this application.

The datasheet shows the dynamic performance as:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=646769;image)

This is really slow, 40 to 100us to turn on, 70-170us to turn off, compared to a normal, unprotected, MOSFET.

This graph shows the transfer characteristics:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=646775;image)

From this graph I can get the transconductance, there is a 20A change in Id for a 2V change in Vgs so the transconductance is 10.

This protected device may be good for static loads, or slowly changing loads. It is not well suited for a dynamic load.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Mechatrommer** on **February 08, 2019, 12:43:53 am**

Quote from: Jay_Diddy_B on February 07, 2019, 11:35:09 pm

> This is really slow, 40 to 100us to turn on, 70-170us to turn off, compared to a normal, unprotected, MOSFET.

according to him, after tuning, the settling time is "fast" 20us (i'm not sure how he get 20us settling time from a 40us rise time device) but well, isnt that a good thing? slower settling/rise time means less overshoot, C11 probably helps in that respect as well. as OP circuit you made is also somewhere if not slower than 20us settling/rise time (not sure if you've made faster latest circuit in the middle of this thread), there is another load circuit, also reporting somewhere 20us settling time but faster ~2us rise time, but that has more overshoot than yours. i guess we only can pick 2 from those 3. anyway jrsikken sold his earlier unit (probably oscillated) to hundred of happy customers already ::)

Title: **Re: Dynamic Electronic Load Project**
Post by: **jrsikken** on **February 08, 2019, 08:12:23 am**

Hi, I quit selling my electronic load because last year sales dropped dramatically, I think because people prefer cheap chinese electronic load. For just $18 you get one with higher power rating, a display, a dial and with graphical userinterface on the pc. I tested one myself and it is pretty cool. The chinese electronic load does not support pulsed loads yet, but it won't take long. I also quit selling because assembly and testing/calibrating took a lot of time. I think it was a wise decision.

Title: **Re: Dynamic Electronic Load Project**
Post by: **MarkF** on **June 15, 2019, 03:30:10 am**

Has there been any issues with @jrsikken circuit spiking to max current when the Device Under Test (DUT) is turned on after the load power is applied?

The basic issue:

When the power is turned on and there is NO load attached there will be no voltage across the R_{SENSE} and any small voltage on the (+) opamp input (or opamp input offset voltage) will cause the output to the MOSFET gate to go to max. Therefore when the DUT is turned on, max current will be pulled until the opamp recovers.

My question comes from this thread and how to address the issue:
https://www.eevblog.com/forum/projects/mosfet-and-opamp-load-tester-constant-current-load/msg2485161/#msg2485161 (https://www.eevblog.com/forum/projects/mosfet-and-opamp-load-tester-constant-current-load/msg2485161/#msg2485161)

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **June 15, 2019, 10:05:11 am**

Hi MarkF and the group,

I am more than happy to answer your question here :D

I am going to illustrate the issue with a simplified schematic of the Dynamic load schematic I presented in this thread.

Model - Normal Operation

If the power source being tested is turned on at the same time or before the load is turned on the current in the load will not overshoot.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763374;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763380;image)

The waveform to look at is the output of the op-amp. The voltage slews to the operating point and the load current rises to the target value.

Model - Load on First

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763386;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763392;image)

If the load is turned on first. The op-amp will be unable to increase the current and the output of the amp will saturate at the positive rail.
When the DUT voltage is applied, the current will overshoot and the op-amp will reduce the gate drive to bring the current to the target value.

Model - large time constants

The model above uses relatively fast loop. At of the load designs found on the internet, do not care about the loop response. They are not interested in measuring the step response of a power supply.
If I populate the circuit with 'slow' components, I get:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763398;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763404;image)

The overshoot is higher and takes longer to recover.

Commercial Load

Commercial loads generally have this feature. If I want to avoid a start up transient I use my HP load in the constant resistance mode. In this mode the current reference is obtained DUT voltage.

The model is attached.

Regards,
Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **June 15, 2019, 10:15:42 am**

Hi,

If the Power supply being tested, DUT, is current limited it will take longer for the load current to come in to regulation:

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763419;image)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763425;image)

Regards,
Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **MarkF** on **June 15, 2019, 10:58:54 am**

Thanks Jay.
My only interest in common with @jrsikken circuit and mine is the single power supply architecture.
https://www.eevblog.com/forum/beginners/a-load-off-my-mnd/msg2151265/#msg2151265 (https://www.eevblog.com/forum/beginners/a-load-off-my-mnd/msg2151265/#msg2151265)

I'm looking at the power up sequence where the load is turned ON before the DUT is turned ON and the current overshoot. I looks like both our circuits show the same issue.

I want to propose that a resistor be added between the R_{SENSE} high side and the positive op-amp supply rail. This would always provide a minimal current across R_{SENSE}. Therefore when the load is set a 0A, the op-amp would always try to turn OFF the MOSFET even if no load is attached. This would prevent a turn ON overshoot. I'm thinking a value between 10K and 100K to provide a minimal current.

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763467;image)

Title: **Re: Dynamic Electronic Load Project**
Post by: **t1d** on **June 15, 2019, 11:02:57 am**

My sincere apologies, Jay. I thought this was on my thread, as it looks very similar. I am embarrassed... So much for not being able to sleep and being on the web at 4 am.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **June 16, 2019, 04:16:19 am**

Quote from: MarkF on June 15, 2019, 10:58:54 am

> Thanks Jay.
> My only interest in common with @jrsikken circuit and mine is the single power supply architecture.
> https://www.eevblog.com/forum/beginners/a-load-off-my-mnd/msg2151265/#msg2151265 (https://www.eevblog.com/forum/beginners/a-load-off-my-mnd/msg2151265/#msg2151265)
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> I'm looking at the power up sequence where the load is turned ON before the DUT is turned ON and the current overshoot. I looks like both our circuits show the same issue.
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>
> (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763467;image)

Hi,

The current introduced by the 10K resistor, you have added, will not be enough to help.

Here is a concept that should work. It essentially modifies the load to be constant resistance at low input voltages. It requires a summing node. The circuit works by amplifying the difference between the input voltage and a reference voltage and reducing the current sense point if the input voltage is low.

Model

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763989;image)
Modelling Results

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=763995;image)

I have attached the LTspice model

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **MarkF** on **June 16, 2019, 01:07:49 pm**

Thanks Jay for taking the time to look at this.
I am going to wire this up and see how it performs.
Having a constant resistance mode would be a nice to have option.

A couple questions:
- I assume the R9 value was chosen to lower the amount of constant resistance influence?
- The R7:R4 ratio is suppose to be a 10:1 voltage divider?

Title: **Re: Dynamic Electronic Load Project**
Post by: **lordvader88** on **June 26, 2019, 06:35:32 pm**

Ok I too shall try this as mine is dangerous on startup with a LM358 and IRF540. I haven't tried changing the op-amp, I have LT1013 too.

Title: **Re: Dynamic Electronic Load Project**
Post by: **MarkF** on **October 05, 2019, 03:30:37 pm**

I finally got around to working on this...
To review the issue:

Initial conditions:
- With the DUT disconnected or powered off
- The control voltage ($V_{control}$) set to 0V.

Resulting in:
- The op-amp drives the MOSFET gate to 11V (max voltage)
- The MOSFET is fully turned on
- When DUT is connected and turned ON, maximum current is initially drawn

Solution:
- Added $R_{min\text{-}set}$ to supply a minimum current (10mA) through R_{sense}
  (My initial $R_{min\text{-}set}$ of 10K was too high. A value of 1.2K seems a good compromise)
- This forces a minimum voltage across R_{sense}

- This artificially raises $V_{control}$ to start turning ON the MOSFET
- The MOSFET gate voltage now goes to 0V for $V_{control}$ less than approx. 10mV

Possible issues:
- When using external voltage control, the actual desired load current will be slightly lower than what is expected.

[attachimg=1 width=1000]

Title: **Re: Dynamic Electronic Load Project**
Post by: **MarkF** on **October 05, 2019, 03:39:54 pm**

Jay,
Maybe you don't see this issue since your control is via the (-) op-amp input.
Where my control is via the (+) input.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Kleinstein** on **October 05, 2019, 05:18:40 pm**

Starting with the gate voltage at a high value, because the DUT can not deliver the commanded current (e.g. DUT open) is a known problem. The main cause is that the control loop does not know about the limitation of the DUT and thus starts with a very high gate voltage it than need quite some time to recover, as the capacitor in the regulator part is charges quite high.

A solution to this is to limit the gate voltage to a low value, if the voltage at the MOSFET drain is too low (e.g. below some 100 mV or so).

A minimum solution is to limit the maximum gate voltage to a reasonable value, just high enough to deliver the maximum current. This would limit the initial current peak, though not fully prevent it.

Adding current to the shunt only adds a small offset. This may solve the problem of having an OP offset that does not allow to set the current all the way to zero. However it does not help if the initial current is set higher than zero when the DUT is connected.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **October 05, 2019, 10:03:34 pm**

Hi group,

Let me start with a little benchmarking. I will display the test results for an HP 60501A (150W) load module in a 6050A mainframe. The load is set to the 3A range and 1A constant current mode.
The current was measured with a Tektronix TCP202A DC current probe.
The power supply used for the testing HP665A set for 5V and 2A limit.

Test 1

[attachimg=2]

This is using the ON/OFF button on the power supply.

Test 2

[attachimg=1]

This is 'hot-plugging' the power supply into the load.

Note: the current scaling was changed from 200mA to 500mA/div

It would be interesting to see the results for testing other popular electronic loads for example B&K, Rigol, Maynuo etc.. I don't have any of these.

Regards,
Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **October 05, 2019, 10:11:42 pm**

Hi group,

Let me do the same tests on the unit that I shared in the first post of this thread.

Link: https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/ (https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/)

Test 1

[attachimg=3]

This is turning using the power supply on/off button.

Test 2

[attachimg=1]

This is 'hot plugging'. Turning the supply on first before connecting the load.

[attachimg=2]

zooming in, you can see the transient is essentially over in 22us.

You can decide if this a problem that needs fixing.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **ogden** on **November 03, 2019, 10:38:14 pm**

I would say that for sake of circuit simplicity it may stay as it is because "overshoot problem" supposedly can be avoided by using power switch of the load - by powering it on only *after* connection to the DUT.

Title: **Re: Dynamic Electronic Load Project**
Post by: **ignilux** on **February 19, 2020, 04:16:41 am**

I realize that this thread is now almost 6 years old, but I came across it again today and have spent some time playing with the excellent LTspice simulations provided by Jay_Diddy_B on the first page. I understand how the circuit as a whole and the individual blocks work, but I had a few questions for Jay and/or others.

First, I'm curious how the resistor values were determined on the grounded non-inverting inputs. Assuming that they're present for offset compensation, the method I'm familiar with says that the resistor should be equal to the parallel combination of the input and feedback resistances. However, the inverting summing amplifier, for example would require a different value depending on the position of the potentiometers, wouldn't it? How are things complicated by the presence of an active device in the feedback loop, as in the two error amplifiers driving the FETs?

Next, I'm wondering how the lead inductance damping network was chosen. The peak in the bode plot is at 200 kHz, and it looks like the cutoff frequency of the damping network is around 33 kHz. Was it simply a matter of picking something with a sufficiently low frequency as to roll the high frequency gain off before the peak, but high enough that the transient response didn't suffer?

Finally, I'm unsure how to interpret the results of some modifications that I made to the model. I've swapped the opamps and FETs, and adjusted the frequency compensation to get rid of some ringing on the rising edge of the load current waveform. However, despite having 80 degrees of phase margin and a gain margin of 20 dB, I'm still seeing 5% overshoot on the rising edges. The AC analysis would seem to indicate a rock-solid control loop, but the ringing gives me pause. What am I missing?

Thanks again to Jay_Diddy_B for being the in-house expert of spice, and for all the work he puts in to threads like these.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **February 20, 2020, 02:05:53 am**

Ignilux and the group,

Thank you for your kind words.

Quote from: ignilux on February 19, 2020, 04:16:41 am

> First, I'm curious how the resistor values were determined on the grounded non-inverting inputs. Assuming that they're present for offset compensation, the method I'm familiar with says that the resistor should be equal to the parallel combination of the input and feedback resistances. However, the inverting summing amplifier, for example would require a different value depending on the position of the potentiometers, wouldn't it? How are things complicated by the presence of an active device in the feedback loop, as in the two error amplifiers driving the FETs?

The general idea of having equal resistance on the non-inverting and inverting inputs is so that any bias currents in the op-amp input will not create an (additional) offset.

If you take the LM324, from the TI datasheet worst case input bias current is 500nA, over the full temperature range.
With 100k \$\Omega\$ of feedback resistance this is converted to 50mV

Consider this model

[attachimg=3]

If you run this model and plot the results:

[attachimg=1]

The potentiometers can be converted to a voltage source and the Thevenin equivalent resistance. The Thevenin resistance will vary with the position of the potentiometer. The maximum resistance is when the wiper is in the middle. At this point the Thevenin resistance is half the value of the potentiometer 10k \$\Omega\$ / 2 = 5k \$\Omega\$

Since the current is set by adjusting the potentiometer, the error introduced by the potentiometer doesn't matter.

Regards,

Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **ignilux** on **February 20, 2020, 02:44:14 am**

Aha! That's both a very clever, and surprisingly simple solution to the problem at hand. Why guess when you can *measure?*

Title: **Re: Dynamic Electronic Load Project**
Post by: **JaysonJT** on **August 19, 2020, 04:22:08 pm**

Hi,

In your constant current Mosfet model, may I know how you picked the 300p and 1000p capacitor values? Also, the gain of 3?

If Vds is 5V, I would have thought you'd pick the values on the red line as shown in the graph below:
(http://)

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **August 20, 2020, 12:11:45 am**

Quote from: JaysonJT on August 19, 2020, 04:22:08 pm

> Hi,
>
> In your constant current Mosfet model, may I know how you picked the 300p and 1000p capacitor values? Also, the gain of 3?
>
> If Vds is 5V, I would have thought you'd pick the values on the red line as shown in the graph below:

JaysonJT,

Welcome to the forum!!

So, you want me to remember what I was thinking in June 2014? ;)

(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=97849;image)

In this example I have increased the transconductance to 6.

Background

There are a tremendous number of load designs on the internet and almost as many questions on how stop a load from oscillating or my load oscillates when I increase the current.

This series of posts was 'engineered' to illustrate how to design a stable load and explain these effects.

When I used the MOSFET small-signal model with a transconductance and the two capacitors I was demonstrating that the ac performance was dominated be these parameters.

The real stability checks were done with the full MOSFET model.

Loop design

Control loops are normally designed with a phase margin greater than 45 degrees and again margin of at least 6dB.

There is a lot room for component tolerances with these margins.

Since the load could be used with any input from 0.5V to VDS rating of the MOSFET you should check stability at the worst case operating point.

In one of the messages I illustrated that if the loop is marginally, the control loop will oscillate if the load current is increased.

Regards,
Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **zeo0d** on **September 15, 2020, 09:29:37 am**

Hi Jay_Diddy_B,

i'm trying to design a pulsed load for different power LED configurations (Current form 0-10A and Voltage from 0-200V), and i was going through the post and think that my problem was similar to what you designed.

But i will use DAC (connected to opamp non-inverting input) for setting the current and voltage.

What i don't understand is: if i choose TPH5200FNH mosfet for my application how can i initially determine before simulation what OPAMP is most suited regarding to GWB, SR, output current capability, single or dual rail, settling time, VF opamp or CF opamp ?

i think i have to calculate the effective input capacitance of the mosfet from Qg total in order do determine what opapm do i need ? I'm not shore how to do that?

I want to pulse the load with max 1kHz (most of the time is 200Hz) and the rise/fall time of mosfet turn on/off should be no more than 1us

Best Regards.

Title: **Re: Dynamic Electronic Load Project**
Post by: **Jay_Diddy_B** on **September 15, 2020, 12:37:20 pm**

Hi zeo0d,

Welcome to the forum.

This is an old thread that was started in September 2013, it just celebrated its 7th Birthday:

[attachimg=1]
(https://www.eevblog.com/forum/projects/dynamic-electronic-load-project/?action=dlattach;attach=1067242;image)

The thread is about designing a small to medium sized load for testing the transient response response of power supplies.

Your request for a high power circuit (0-10A, 0-200V) to test LED arrays is off-topic.

I suggest that you start a new topic. Indicate why you need the circuit and what you are hoping to measure with.

Regards,
Jay_Diddy_B

Title: **Re: Dynamic Electronic Load Project**
Post by: **Vovk_Z** on **September 15, 2020, 12:59:28 pm**

Quote from: zeo0d on September 15, 2020, 09:29:37 am

> What i don't understand is: if i choose TPH5200FNH mosfet for my application how can i initially determine before simulation what OPAMP is most suited regarding to GWB, SR, output current capability, single or dual rail, settling time, VF opamp or CF opamp ?

I'm not sure about TPH5200FNH, it is possibly too weak even for very short duration of measurement. You have to carefully look at it's SOA, and don't work very close to the SOA border.
About opamp: I'l use some similar to was used in other fast electronic loads. There are one or two designs published in the internet. Or otherwise something like OP27/OPA189 may work (they are the fastest easy obtainable and still cheap precision opamps I know).
We usually use VF opamps here. CF opamps are typically faster but have lower accuracy (higher offset etc). You may use CF opamp if you **really** need much more speed then any VF opamp can give.

Title: **Re: Dynamic Electronic Load Project**
Post by: **blackdog** on **October 04, 2021, 04:13:06 pm**

Hi Jay_Diddy_B,

I was once again going through some topics that deal with "Electronic Loads"
As far as I can tell, your explanation is excellent!

As I myself use a faster Electronic Load(Jim Williams Design) for some tests, and you have also fitted your design with a faster opamp but that did not bring much faster edges.
Which I therefore think is too little current available to drive the Gate impedance fast enough.

Would it make sense to use e.g. an LT1010 as a buffer with a faster opamp to get a real jump in the edge speed?
The opamp used does not see a load when using the LT1010 and the LT1010 has a good drive capability for capacitive loads with its 7 Ohm output resistor.
Then to make it optimal, the gate resistance can be lowered a bit because the LT1010 already takes care of this for a large part.
A nice opamp with a lot of phase margin is the ADA4625 and this one also has inputs that includes V-
I am aware that these parts cost a bit more, but going out for dinner with your family costs a lot more and you usually forget about it after a few weeks :-)

Is it possible that you could test this once in LTSpice if you have the time?
My skills with LTSpice are not very good yet.
Building a test circuit now would be even faster for me :-)

I use this Active Load when I need fast edges.
(http://www.bramcam.nl/AL/JW-BK-DLT-01.JPG)

Here is the above diagram in a box.
The Active Load is only used for speed testing and not for long time dissipating for much power.
I can mount a heatsink against the bottom as needed to dissipate some more.
(http://www.bramcam.nl/AL/45-Active-Load-Wiliams.jpg)

For anyone working with fast or basically all Active Loads, twist your cabling between the source to be tested and the Active Load.
With the Jim Williams version as I showed above, every mm of wiring between the Active Load and the source is suspect if you want neat edges.
For anyone working with fast or basically all Active Loads, twist your cabling between the source to be tested and the Active Load.
With the Jim Williams version as I showed above, every mm of wiring between the Active Load and the source is suspect if you want neat flanks.

Thank and kind regads,
Bram

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